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Robotic leaping enhanced by thrust-induced hypogravity, achieving precise, predictable, and extended jumps

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Robotic jumping research advances engineering and biomimicry frontiers, prioritizing range, precision, and predictability to navigate unstructured environments. Earth's gravity necessitates powerful actuators and lightweight bodies in robotic designs for maximal jump height. While many robots excel in statical environments, precise, predictable jumps in dynamic settings remain challenging. We realized this with a bipedal robot leveraging thrust-induced hypogravity, alongside dual regulation of aerial attitude and parabolic trajectory via thrust vectoring. Hypogravity multiplies leap range (max: 6.9 m) despite leg force saturation, enabling the robot to clear multi-level stairs, a 2.35-m-high wall, and 3-m-wide stream. Parabolic trajectory regulation allows leap distance precision/consistency surpassing existing thrust-assisted hybrids and leg-only jumpers. It enables pre-jump prediction of aerial/landing positions and timing, facilitating leaps in dynamic scenarios: through fast-moving windows (3.8 m/s), onto shifting, confined targets, and against wind disturbance. This research establishes extended range, precise, and predictable jumping through self-generated hypogravity and parabolic trajectory regulation.

Jumping locomotion, a biomechanically efficient strategy for surmounting large obstacles and enabling rapid traversal across unstructured terrains, is extensively adopted in nature. Jumping specialists in animal kingdom can achieve superior jumping ranges, cope with dynamic environmental uncertainties via predictive jumping, and stably land onto confined target areas. However, for robotics, the pursuit of extended leap range also leads to an inherent compromise of leap precision, predictability, landing stability and safety. Conquering this trade-off to reach animal-level jumping performance with trajectory predictability, landing precision and stability, and ultra-broad range remains a key challenge. Paradigms for maximizing jumping elevation frequently involve enhancing the power-to-weight ratio, entailing advancements in jumping actuation systems. A prevalent approach is the use of catapult mechanisms, such as loading springs with cam mechanisms[1–5], winching mechanisms[6–15], and mini-scale jumpers with bistable structures adopting torque reversal mechanisms[16–21](also commonly adopted by robotic grippers[22–24]), soft buckling beam[25,26], or shell[27–30] structures. Nonetheless, catapult-based jumping's energy-storing phases, often much longer than the leap aerial phases, hinder agility and continuity. Its reliance on elastic energy poses difficulties for precise feedback control of trajectories or landing positions, most such jumpers are manually, or open loop controlled. In contrast, actuator direct-drive methods (e.g., motor[31–33], electrohydraulic actuation[34,35]) enable agile, continuous jumping with better controllability. Still, their limited instantaneous power restricts ultra-high leaps (e.g., 3 m high), with the highest record (1.25 m) achieved by Salto-1P[36].

Previous works have employed fixed wings as aerial assistance for elongated aerial phases[7,37–40]. While these systems cover enhanced leap distances, the added wing structures also increase overall system mass and vertical air drag, impairing jumping elevation. Meanwhile, the aerodynamic forces generated by wings also interfere with ballistic

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leap trajectories, causing difficulties in aerial path and landing position prediction. Truong et al.[41]. used vertical flapping wings combined with catapult launch mechanism, enhancing jump height by about 30%. However, its maximum height remains restrained under 1 meter, and its cyclical flapping forces distort the trajectory's parabolic feature, introducing both trajectory and landing position unpredictability.

The vast majority of robotic jumping or hopping research is under the ubiquitous constraint of Earth's gravity[1–21,25–40,42–46]. There is a lack of research unveiling the potential of thrust-induced hypogravity (using air thrust to offset Earth's gravity) for jumping motions. Our rationale for studying hypogravity jumping is multifold:(1) Leap range enhancement. Linear gravity reduction shows a superior enhancement effect (hyperbolic) on jumping elevation compared to linear launch velocity increase (quadratic). Theoretical height can achieve infinity with the gravity fully offset; (2) Landing safety. Reduced landing speed (close to take-off speed) ensures deducted landing impact; (3) Energy efficiency. Compared to direct flying, legged propulsion force instantly establishes a high initial speed, only requiring a constant thrust below robot's weight, while flight demands constant upward acceleration (thrust ≥ weight). (4) Alternative ways for parabolic trajectory modulation. Jumping height gain ($h = v_{takeoff}{}^2/2g^*$) can be tuned via the gravity constant$\hat{g}$.

In literature, the precise control of parabolic leap trajectory and landing position is a challenging issue. For leap trajectory control, current state-of-the-art robotic jumpers mainly utilize the delicate control of launch speed and angle (e.g., Salto-1P[47] (2020), Moobot[19]), showcasing effective tracking of individual parabolas, or regulate touchdown attitude (e.g., Salto-1P[48] (2018), Hopcopter[49], flapping-wing passive hopper[50]) for overall hopping path control. All these works lack parabolic trajectory regulation during mid-flight. In real-world applications, terrain surface variations could easily cause the deviation of launch or touchdown angle, leading to large trajectory errors (e.g., In a chair-desk jumping task, Salto-1P[48] has a strict 0.7-degree touchdown angle error tolerance). Also, extreme cases such as wind disturbances could distort a trajectory's ballistic pattern, rendering such methods insufficient. For robustness and predictability of leap patterns, we investigate a previously underexplored locomotion pattern: parabolic trajectory regulation under thrust-induced hypogravity condition. It involves studying the discrete-point errors in every timestamp of a full leap parabola, under the guidance from a reference parabolic target with coupled time and positions (time-coupled parabolic trajectories, TCPT). Previous fixed-thrust-based jumping hybrids (thrust direction fixed to airframe) employed active thrust propulsion for touchdown attitude and height regulation[49,50], non-parabolic position control[51], or horizontal velocity and height control[52], to initiate rebound-based motions. These systems exhibit limited precision in regulating individual parabolic trajectories, not only because of their control objectives but the inherent fixed-thrust configuration limitations. Specifically, attitude adjustments require airframe rotation that induces extra translations, while positional control rigidly requires airframe rotation, restricting the control flexibility of inclination. Another technical contribution of this work lies in decoupling the rigidity between the regulation of inclination and parabolic trajectory and enables simultaneous control of the two components.

Following the mid-flight regulation of leap trajectory, balanced landing without secondary bounces is critical for tasks involving confined targets. Many jumpers show imbalance upon landing, hence using self-righting mechanism to recover from instability[5,6,8–10,53–55]. Soft-bodied jumpers[27,29,30,34,56–61] can partly reduce sary bounces with inherent compliance. Yet, balanced and bounce-free landing (stop at the first land point) is insufficient in the literature.

To tackle the aforementioned challenges, we devise a 370 g bipedal robot called GravOff (Gravity offset) robot (Fig. 1A) that first investigates robotic jumping under thrust-induced hypogravity while regulating the aerial-phase parabolic trajectories. The component of hypogravity endows the robot with range-tunable projectile motions, extending leap ranges multifold despite the saturation of leg joint torque. This enables the robot to achieve a height jump up to 46 times body height (6.9 m), and facilitates large, rapid spatial transitions in field tests: leaping over multi-level stairs, a 2.35 m-high wall, a 3 m-wide stream, etc. An energy comparison study between hypogravity jumping and trajectory-controlled flight modes reveals that the total energy cost of hypogravity jumping modes is less than 10 % of that of the flight mode when overcoming the same medium-scale obstacle of ~5 body heights. The legs cooperate with thrusters to control directional jumping motions, enabling both longitudinal and lateral leaps. During aerial descent, the legs act as an active deceleration mechanism, dissipating 50% of the impact momentum compared to direct landings, thereby facilitating bounce-free landings within confined targets. Legs' propulsion and active deceleration are coordinated with vector-thrust and attitude control to enable seamless consecutive leaps in challenging scenarios, including jumps across inclination-varied targets and across area-confined, dynamic targets. With directed takeoff, the robot leaps under thrust-induced hypogravity while regulating the projectile motion for aerial-phase and landing-point precision and predictability. We use synchronous thrust vectoring to enable dual regulation of inclination and parabolic trajectory. The system reveals effectiveness in targeted jumping and shows an order of jumping distance (JD) precision and consistency improvement than our fixed-thrust-based regulation case and previous precise leg-only jumpers, which could achieve landings within a robot-projection-sized target region (0.25 m by 0.25 m) with sub-centimeter distance error and standard deviation. Under interference, such as large takeoff angle deviations (up to 15°) and wind disturbance (3.5 m/s), the robot can remain targeted landing with small variations. Owing to the time-coupled feature of the reference parabola (i.e., the TCPT), the timing for all positions in the subsequent parabolic curve, including the landing point, can be estimated. This allows the robot to execute well-timed leaps in response to dynamic environmental changes. The robot demonstrates this capability by conquering rapid-moving obstacles, such as leaping through a pendular window at 3.8 m/s and a rotating window at 100 deg/s. Combined with jumping, the robot also conducts versatile motions in field tests, including legged crawling (on grassland, sand, cobbles, etc.) and air-propelled wheeling (reaching 24 body length/s).

Results

Prototyping the GravOff bipedal robot

Bush crickets (*Tettigoniidae*) have long and thin legs, but can jump in tens of milliseconds. Their elongated limbs facilitate an extended acceleration phase while allowing for reduced muscle force demand to achieve high take-off velocities. They power their jumps by muscle direct actuation that exert forces on long levers and showcase superior quickness than spring-actuated jumpers (such as locust) for escape movements[62]. The GravOff robot also utilizes direct actuation method, with a pair of 3-Degree-of-Freedom (DOF) long legs (leg extension ratio reaches 340%) to initiate jumps resembling bush crickets (Fig. 1B, C), experiencing balanced crouching state, synchronous leg extension, and the airborne phase. For the proposed hypogravity effect, the four thrusters are controlled at a constant rotational speed while balancing the robot simultaneously (by a cascaded Proportional-Integral-Derivative (PID) attitude controller, see Fig. S2 and Table S2). The thrusters are arranged with a relatively wide span (220 mm by 190 mm) for long moment arms to facilitate balancing stability and aerial attitude control.

The 2-DOF jumping unit (Fig. 1D(i)) features a parallelogram mechanism and is reinforced with two additional bars for increased lateral stiffness. A brushless DC motor (Fig. 1D(ii)) serves as the primary power source, driving the upper link of the mechanism. The second DOF, controlled by servo 1, manages the overall rotation of the jumping unit. We adopt this integrated design for the jumping unit to

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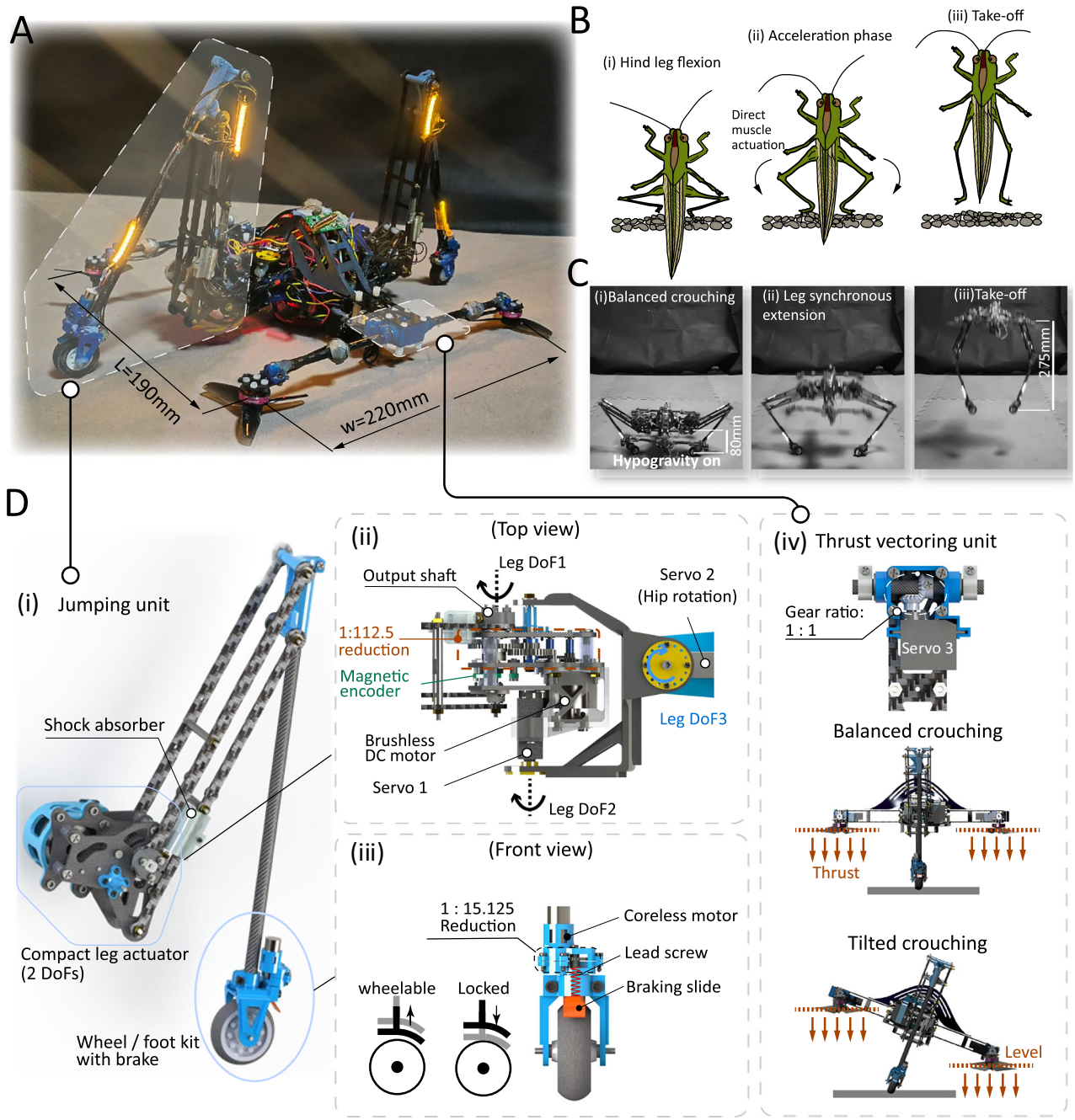


Fig. 1 | Prototyping the GravOff bipedal robot. A Overview of the GravOff robot in sitting pose. The robot has four DOF in each leg (two for jumping initiation, one for hip rotation, and one for the braking of the passive wheeling foot) and four thrusters for gravity offset and aerial trajectory regulation. **B** Demonstration of a vertical jump by bush crickets (Tettigoniidae). Bush crickets leverage direct muscle actuation and feature a pair of long and thin hind legs for a long duration of leap acceleration phase, and reduced muscle force requirement[62]. **C** Vertical jumping of the GravOff robot. The robot achieves leaps via direct motor actuation driving a pair of long legs (leg extension ratio: 340%), while the thrusters maintain hypo-gravity conditions throughout the experiment. **D** Structural details of the GravOff robot. (i). The jumping unit. To minimize the moment of inertia, the jumping unit has its two rotational DOF for jumping embedded in a single compact actuator. The wheelable foot (with brake) is intended for both jumping (brake on) and wheeling (brake off) scenarios. The shock absorber is used for damping impacts upon landing. (ii). Top view of the custom-designed leg actuator. The brushless DC motor provides the major torque for jumping, and servo 1 controls the overall rotation of the leg actuator. (iii). Foot brake based on a micro screw-slider mechanism. (iv). Thrust vectoring unit. This unit controls the pitch angle of thrust and enables the rotation plane of the thrusters to maintain level during the tilted crouching pose.

reduce the moment of inertia and alleviate torque demand for the leg actuator. To mitigate landing impact, a 3D-printed TPU shock absorber is placed between the upper and lower links of the four-bar mechanism. A third DOF is adopted for hip rotation to enable legged assistive crawling and locomotion mode transformation (to wheeling). A lockable wheeling structure featuring a micro screw-slider mechanism (Fig. 1D(iii)) is applied at the foot for both the jumping mode (wheels locked) and a high-speed wheeling mode for flat terrain. A thrust vectoring (TV) unit (see Fig. 1D(iv)) works in conjunction with the thrusters to change the initial launch direction of a jump. It eliminates the horizontal thrust component and prevents foot sliding when the robot starts jumping in the tilted balancing pose. All components used by the robot are listed in Table S1 and the electronics architecture is discussed in the supplementary section S1 and Fig. S1.

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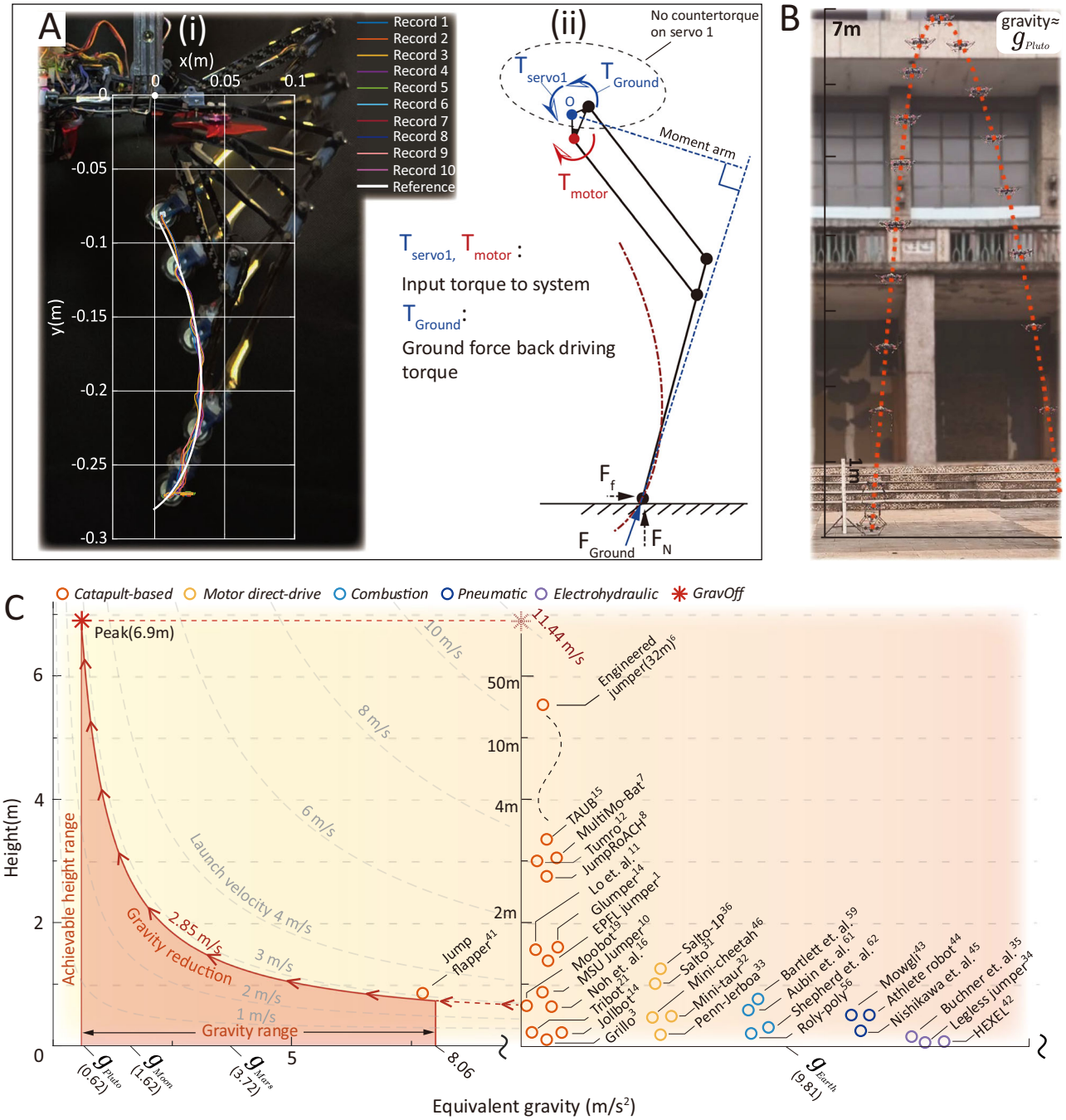


Fig. 2 | Achieving leap extreme via hypogravity. A The leg extension trajectory applied in the jumping motion. (i) The foot tip trajectory generated by the coordinated control of the four-bar mechanism and servo 1. The reference and ten records of actual trajectories (captured by Video analysis tool Tracker) were compared. (ii) Actuator torque analysis. As the foot tracks the assigned trajectory, the ground force back driving torque T_{ground} maintains the same direction as the input torque of servo 1 (T_{servo}). This configuration alleviates the torque demand on the leg servo 1 and allows the jumping performance to primarily depend on the leg motor. **B** Peak leap height test under microgravity. Under a gravity close to Pluto (0.620 m/s^2), the GravOff robot achieved an apex height of 6.9 m (46 times its body height). **C** The achievable height and gravity range of the GravOff robot compared to other types of jumpers. The gravity reduction boosts leap height hyperbolically. With an identical takeoff velocity of 2.85 m/s, it enhances height more than ten-fold compared to Earth's condition.

Leg extension trajectory generation

During jumps, the leg extension trajectory, shown in Fig. 2A, is generated by the coordinated motion of the leg motor and leg servo 1 (In the context, the leg motor refers to the brushless motor in the leg actuator in Fig. 1D(ii)). We applied a slightly convex profile for the foot trajectory by linearly mapping the angle of the leg motor (controlled at 100 deg/s) to leg servo 1, with the two actuators rotating in opposite directions (see supplementary methods in Fig. S3 and section S2). Results show that the recorded trajectories (10 trials) align well with the reference trajectory, with slight deviations from linkage oscillations (see also Supplementary Movie 1). The leg motor is controlled by a bi-directional electronic speed controller (ESC; the control signals and property settings are discussed in supplementary sections S1, S4 and Table S4). The leg motor's position and speed control are based on

two fine-tuned clamped P controllers (see supplementary section S3, Fig. S5).

There exists a difference in torque demand for the two DOF in the jumping unit, as illustrated by Fig. 2B(ii). While the legs exert forces on the ground, the reactive force produces a back driving torque (counterclockwise) on the actuators. This torque aligns with the direction of the servo 1 input torque and opposes that of the leg motor. This design reduces the torque demand on servo 1, requiring it only to regulate the leg trajectory while the leg motor handles the primary torque generation for jumping. Result also reveals that this leg mechanism can provide sufficient torque demand for vertical jumps more than five times of the robot's body heights, without the gravity reduction (thrusters' actuation), as presented in supplementary Fig. S10, section S8 and Supplementary Movie 2 (clip 1). Still, the leap range was shown to be markedly enhanced with constant gravity reduction in comparison to the case of leg-only jumping (see Fig. S11 and clip 2 in Supplementary Movie 2).

Achieving leap extreme via hypogravity

We tested the peak leap height by reducing gravity to microgravity level. As shown in Fig. 2B and Supplementary Movie 3, the GravOff robot attained an experimental peak leap height of 6.9 meters (46 times body height) near g_{Pluto}(0.620 m/s^2), a record surpassing most jumpers working under Earth gravity (Fig. 2C). The robot achieved this with an estimated takeoff velocity of 2.85 m/s (derived by $g^* \Delta T_{aerial}/2$). By comparison, achieving equivalent height under Earth gravity requires 11.62 m/s—a velocity exceeding typical power limitations of legged propulsion systems. From a landing safety perspective, the reduced takeoff velocity in constant hypogravity lowers landing impact momentum to just 25% of an equivalent Earth-gravity jump, thereby reducing damage susceptibility. Overall, the robot demonstrates a broad achievable hypogravity and height range, as depicted in the red shaded area in Fig. 2C, with g^* ranges from 0.062 to 0.822g_{Earth}(Here, 0.822 g_{Earth} corresponds to 10% threshold throttle ensuring stable balancing), and h ranges from 1 to 46 times body height. In the test, the equivalent gravity g^* was set by referring to the quasi-linear gravity-throttle relationship presented in supplementary section S7 and Fig. S9.

Energetics study of the hypogravity jumping

Under thrust-induced hypogravity, jumping elevation is determined by the combined contribution of the legged propulsion and gravity reduction. While hypogravity reduces the energy demand on the leg mechanism, generating the thrust itself requires energy expenditure. To investigate the energetics of hypogravity jumping, we included a statistical test with control variate method to study the total energy cost required to achieve a fixed jump height (0.8 m) under varying levels of equivalent gravity g^*. Eight groups of tests were conducted for the energy analysis, with g^* ranging from 0.1 g to 0.8 g. For each group, the target value of equivalent gravity (g^*) was set using the thrust-throttle mapping in Fig. S9, and leg extension speed was adjusted to achieve the target apex height (0.8 m; as the controlled variable) within ±0.05 m, as validated in Fig. 3A. The results demonstrate a significant reduction in total energy consumption with increasing hypogravity (g^*), decreasing from 242.68 J at g^*=0.1 to 49.52 J at g^*=0.8. Conversely, gravity reduction substantially lowers the leg energy demand (calculated by $E_{leg} = E_{total} - P_{g^*} \cdot \Delta T_{aerial}$, P_{g^*} is the power at g^*, detailed in Table S5) required to achieve a fixed jump height (0.8 m), from 27.09 J at g^*=0.8 to 5.37 J at g^*=0.1. The power is calculated by the product of smoothed voltage and current data (measured by the voltage and current sensors presented in Fig. S1). Detailed data for all trials are provided in Fig. S6 and Table S5-S8.

Our study further demonstrates that hypogravity jumping constitutes a more energy-efficient strategy than flight for clearing medium-scale obstacles (2-8 BH). This was further validated through direct energy comparisons between trajectory-controlled flight and hypogravity leaps, when clearing a 0.73 m table, as shown in Fig. 3C–E and Supplementary Movie 4. The flight was commanded with a reference parabola with a nominal takeoff condition of $v_0^* = 0.6m/s, \phi_0^* = -13^\circ, g^* = 0.018g$, controlled by the fixed-rotor (FR) based MPC (see details in Methods), while gravity-offset leaps employed open-loop pre-tilt launches with $\phi_0 = -6^\circ$. The comparison reveals that hypogravity jumps require a total energy demand <10% and average power demand around 40% (56.28 W versus 139.14 W) of the flight mode. Even using the lighter, legless configuration (193.8 g, Table S1) for a fair comparison, flight still consumes more energy (minimum 266.16 J) than jumping (56.52 J). This energy cost is calculated by multiplying the critical lift power (interpolated from Table S5) by the average flight duration in following the reference flight path (Fig. 3D). For optimal efficiency, the energy feature of hypogravity jumping can guide the strategy of tuning equivalent gravity according to obstacles' sizes. For small-to-medium obstacles (<8 body heights, see illustration in Fig. 3F), leg dominant jumps can be utilized to ensure efficiency (with g^* set to 0.6 g – 0.8 g). For large obstacles where legs propulsion forces saturate, gravity level is reduced accordingly to extend parabolic trajectory range (as exemplified by Fig. 2B).

Longitudinal jumping with the pre-tilt method

To achieve more consistent takeoff angles and velocities, the robot employs the pre-tilt method for longitudinal jumps (see Fig. 4A and Fig. 1D(iv)). During the pre-tilt phase, a 0.6-s ramp-up angle command ϕ_0 is sent from the Teensy controller to the low-level PID attitude controller while the tilt servos (Fig. 1D(iv)) simultaneously rotate to a negative angle $-\phi_0$. Once the pre-tilt stance is achieved and stabilized, the robot applies synchronous extension commands to both legs. This method aligns the ground reaction force closely with the leg extension line, resulting in a consistently directed initial velocity. Owing to the thrust vectoring unit, which eliminates horizontal thrust components, the robot can maintain stability during the pre-tilt phase under oscillatory angle commands, as demonstrated in Supplementary Movie 5. Beyond this pre-tilt-based longitudinal jumping method, the GravOff robot can also initiate lateral jumps by creating phase differences between its leg motors during extension, as shown in Supplementary Section S6, Fig. S8, and Supplementary Movie 6.

Leap trajectory regulation with decoupled inclination control

The precision of parabolic jumping trajectories is regulated after the robot establishes its initial state ($\mathbf{x}_0$ comprising three-dimensional position and velocity) in the jumping-specific coordinate frame $\boldsymbol{Q}_J$ (Fig. 4B; transformed from ground coordinate Q_G, detailed in Methods) via legged propulsion. To address initial state deviations and aerial perturbations, a reference TCPT guides the robot's parabolic motion to ensure trajectory precision and robustness. This TCPT reference includes both trajectory positions and velocities, governed by three tunable parameters: nominal takeoff velocity v_0^*, inclination angle ϕ_0^* (herein also termed pitch), and equivalent gravityg^*:

$$\mathbf{x}_i^{ref} = \begin{bmatrix} x_i^{ref} \\ y_i^{ref} \\ z_i^{ref} \\ u_i^{ref} \\ v_i^{ref} \\ w_i^{ref} \end{bmatrix} = \begin{bmatrix} 0 \\ v_0^* sin(\phi_0^*) t_i + y_0 \\ v_0^* cos(\phi_0^*) t_i - g^* t_i^2/2 + z_0 \\ 0 \\ v_0^* sin(\phi_0^*) \\ v_0^* cos(\phi_0^*) - g^* t_i \end{bmatrix} \quad (1)$$

In each time frame of the aerial phase, the trajectory error $\mathbf{e}_i = \mathbf{x}_i^J - \mathbf{x}_i^{ref}$ is passed as the state input to a time-varying model-based predictive controller (MPC) that regulates motions in the x-y plane and z-direction (see the "Methods" section for details).

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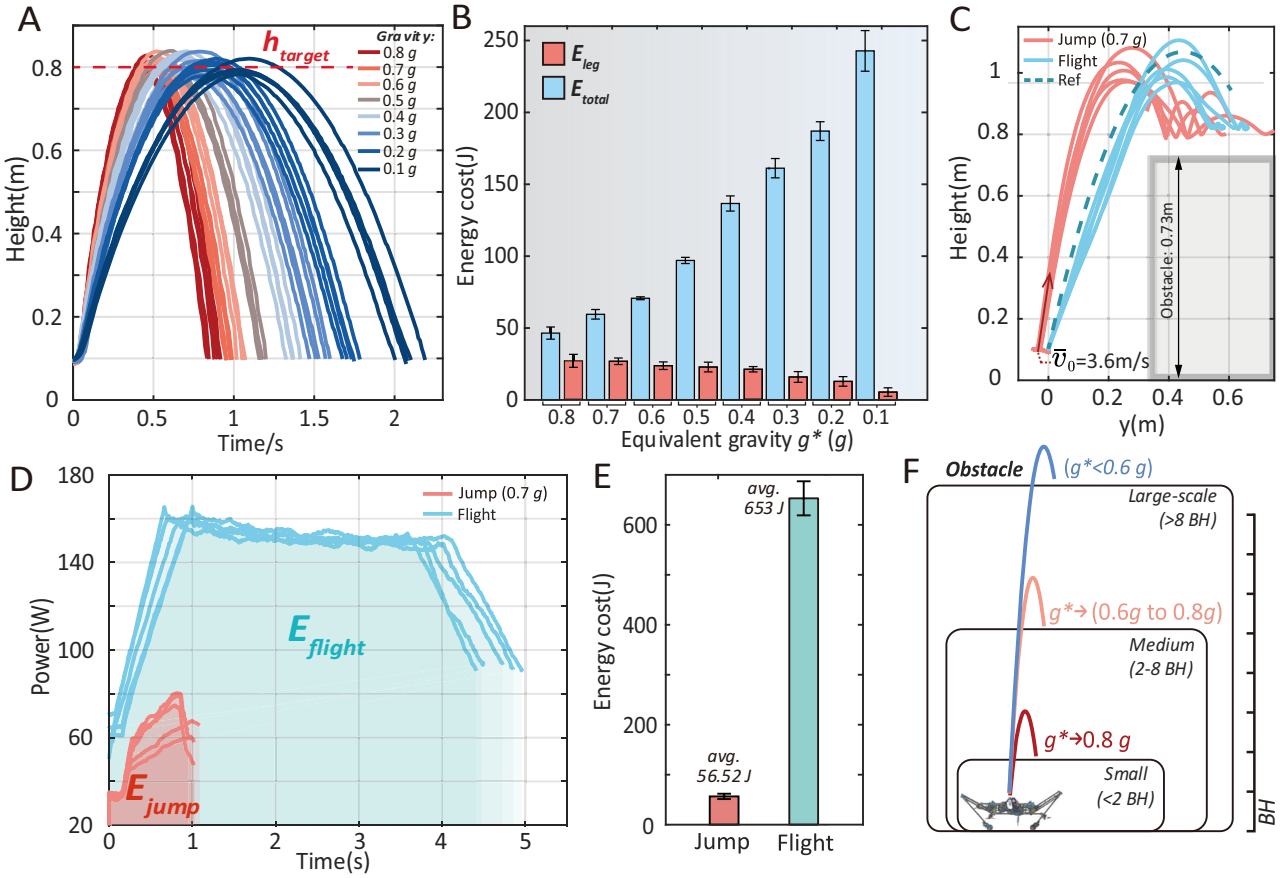


Fig. 3 | Energetics study of the hypogravity jumping. A Leap heights versus time under a range of equivalent gravity levels. The robot leaped and achieved the desired apex elevation (0.8 ± 0.05 m) under eight preset equivalent gravity g^*. The jumping duration extended with the reduction of gravity. **B** Statistics of the total energy cost and the components from the legged propulsion. Total energy consumption drops significantly with the increasement of the equivalent gravity constant, from 242.68 J at g^*=0.1 to 49.52 J at g^*=0.8, while gravity reduction alleviated energy demand from the leg mechanism. The energy component for gravity offset increased with gravity reduction due to both extended duration and greater thrust force requirements. **C** Obstacle clearance of a 0.73 m-high table via hypogravity jumping versus flight. Flight trajectories were controlled by FR-based MPC (see Methods) using a reference parabolic curve under a 0.018 g gravity, with initial velocity of 0.6 m/s and launch angle of −13°. The hypogravity leaps were initiated with g^*=0.7 g, and an average takeoff speed of 3.6 m/s. **D** Power profiles for jumping and flight modes. Power curves represent the product of measured voltage and current, smoothed using MATLAB's smooth function (moving average method, span=0.04). **E** Total energy comparison between flight and jumping modes. Total energy was calculated by integrating power from motion initiation until robot-table contact. **F** Equivalent gravity g^* selection strategy for obstacle clearance. GravOff robot employs a high hypogravity (0.6 g to 0.8 g) to optimize energy efficiency when clearing small-to-medium obstacles (<8 body heights). For large obstacles exceeding leg capability limits, a reduced gravitational constant is set to extend its leap range accordingly.

While standard FR-based dynamics provide sufficient control degrees of freedom for 3D trajectory regulation (roll θ for x-direction, pitch ϕ for y-direction, total thrust ΣT for z-direction), this approach becomes inadequate in jumping scenarios involving terrain surface variations, where the robot must dynamically adjust its body inclination alongside parabolic trajectory regulation to ensure proper contact and stability. However, FR systems require attitude changes to modulate trajectories, i.e., by tilting the entire body to generate horizontal acceleration, which fundamentally couples trajectory control with body inclination.

To decouple inclination control from trajectory regulation, we utilized synchronous thrust vectoring to govern longitudinal trajectory (see comparison in Fig. 4C) and established a thrust-vectoring-based MPC for the discussed jumping scenario (see Methods). The thrust-vectoring-based method assigns controller commands as rotor tilt angle ε_r, roll θ, and total thrust ΣT, while the inclination ϕ_{active}^{cmd} being freely tuned. These commands are passed as reference signals to the low-level attitude controller and tilt servos, as shown in the control workflow (Fig. 4D). Here, the measured attitude angles θ_{mea}, ϕ_{mea}, along with the previous $\varepsilon_r|_{k-1}$ (absent in FR mode), $\sum T|_{k-1}$ are used for state-space model update. Both TV-based MPC and FR-based MPC can regulate the TCPT and can be interchanged during distinct mid-air phases. Concurrently, the leg unit is controlled using current height information in preparation for terrain interaction (active and passive damping, detailed in supplementary section S9).

Leaping onto confined targets

To showcase the performance of the TV-based trajectory regulation, we design a task where the robot leaps onto its projection-sized target region (0.25 m by 0.25 m) under dynamically varying inclination commands, as shown in Fig. 5A and Supplementary Movie 7. The GravOff robot initiated the jump in a start region with a pre-tilt angle of -15° and a desired speed set of 2.0 m/s (by referring to the statistical linear relationship between motor speed and takeoff speed, see Fig. S7B and section S5 for details), followed by the ascending phase where the inclination was maintained, and the descending phase where a step change $\delta\phi = 30^\circ$ lasting around 0.27 s occurs, before landing onto the target region. Figure 5B shows the trajectory reference, result and the TV-based MPC predictions at each time frame, and

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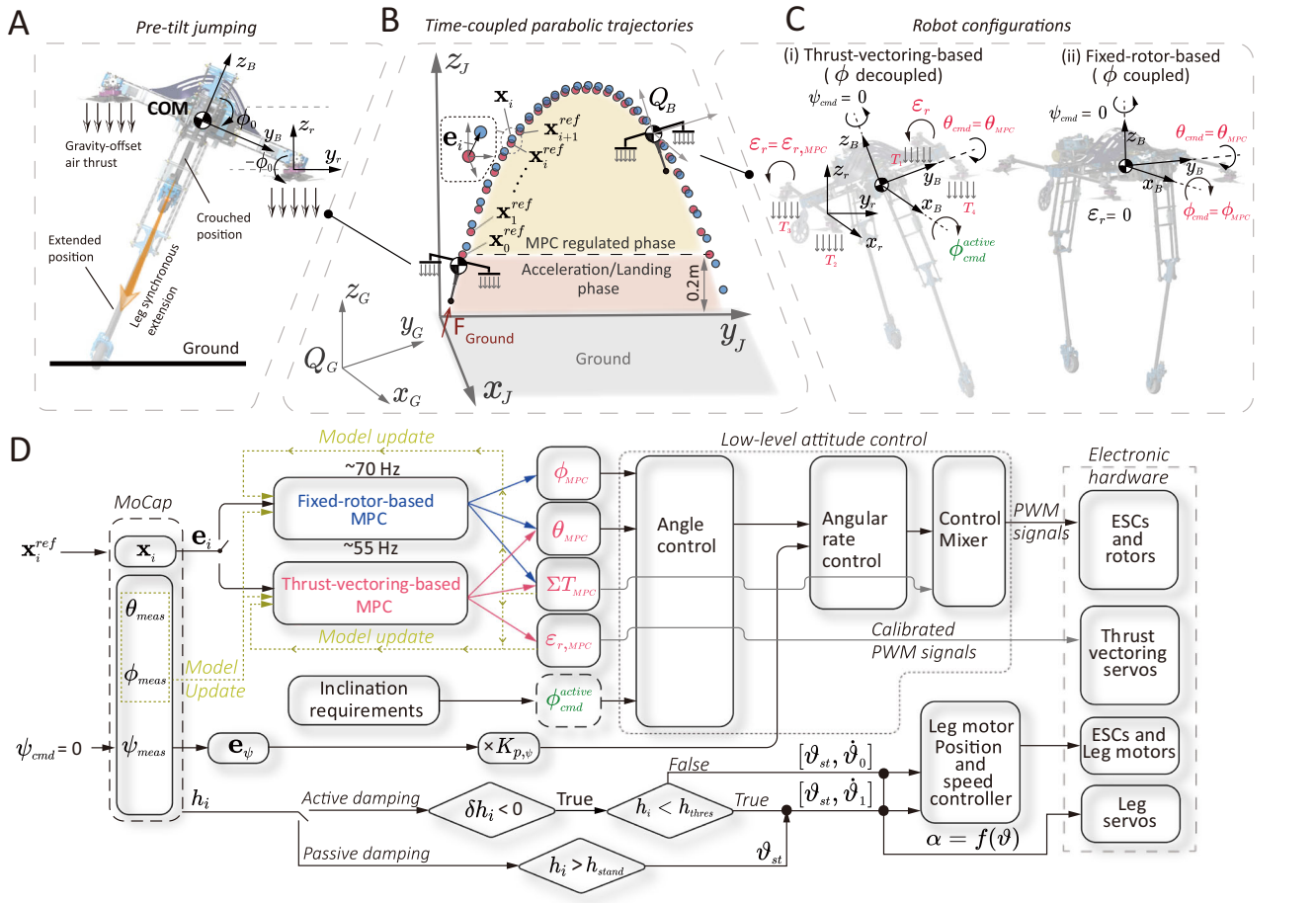


Fig. 4 | Parabolic trajectory regulation with decoupled inclination control. **A** Demonstration of the pre-tilt jumping. Pre-tilt jumping is used to establish the jumping initial states. It aligns the ground reactive force closely with the leg extension line to ensure takeoff angle consistency. **B** Schematic diagram of a TCPT tracking. At each time frame, the tracking reference position is updated according to a theoretical jumping profile defined by the nominal takeoff velocity and gravitational condition (i.e.,v_0^*, ϕ_0^* and g^*), marked as points $\mathbf{x}_i^{ref}$ ($i = 1,2, \ldots, n$). All TCPT references are constructed in the jumping coordinate frame Q_J. The height threshold value of the MPC-regulated region is 0.2 m. **C** The robot's configuration in the aerial phase. The robot utilizes synchronous thrust vectoring (TV) with tilt mechanisms to regulate TCPT in the longitudinal direction and offer control freedom for the inclination angle ϕ. In comparison, free inclination control is unfeasible for fixed-rotor (FR) based configuration as it is coupled with the process of trajectory regulation. In the thrust vectoring mode, the same rotor-fixed refence frame $\boldsymbol{Q}_r$ is used for both the front and rear tilt rotor pairs. **D** Control workflow for the parabolic trajectory regulation. Trajectory regulation is realized by independently relying on the TV-based or FR-based MPC or switching in between. In every loop, the motion capture system's measured states $\mathbf{x}_i$ are passed as inputs to the MPCs, and the measured attitude angles ϕ_{mea}, θ_{mea}, and the previous tilt angle $\varepsilon_r|_{k\text{-}1}$ (FR-based MPC eliminates this) and thrust command $\Sigma T|_{k\text{-}1}$ update the TV-based MPC. The attitude commands from MPCs (with $\psi_{cmd} = 0$) are passed as inputs to the low-level attitude controller that controls the rotors, and the thrust tilt command $\varepsilon_{r,MPC}$ is sent to the servos through calibrated PWM signals. In the aerial phase, the legs are also concurrently controlled according to the current height information to prepare for active damping or passive damping upon landing (see section S9 for details).

Fig. 5C displays the robot's attitude angles, rotor tilt command, and resultant thrust command. The results reveal that the thrust vectoring command ε_r smoothly adapts to the variation of robot's inclination (Fig. 5C(i), (ii)), maintaining the resultant thrust quasi-vertically upward while adjusting longitudinal trajectories, as ε_r takes $-\phi$ as the its reference angle (see details in "Methods" section). The predictions are updated at each time frame, projecting trajectories ten steps ahead. Figure 5D details the absolute prediction errors for each step in every time frame 10 steps ahead of the end time (for the jump presented in Fig. 5A).

To statistically evaluate how well the predictive controller captures the dominant dynamics of the robotic system and the precision of the parabolic path regulation, we conducted repeatability tests for three cases: TV mode with varying ϕ, TV mode with fixed ϕ, and FR mode. Each case was tested in the targeted leaping task (Fig. 5A) across five repeated trials. We assessed the statistical performance of the predictions in three directions using the prediction mean absolute error (PMAE), calculated by:

$$PMAE_i(k) = \frac{1}{p_{stps}} \sum_{j=1}^{p_{stps}} \left| x_{(i)}(k+j|k) - x_{(i)}(k+j) \right|,$$
$$for\ i \in \{x, y, z\}, k = 1, \ldots, n,$$
$$and, \begin{cases} p_{stps} = p, k < n - 10 \\ p_{stps} = p - n + k, else \end{cases}. \quad (2)$$

Here n and p are the total number of time steps and prediction horizon, respectively. Results in Fig. 5E reveal that the experimental trajectories align well with the parabolic references, with the mean absolute prediction error constrained below 0.05 m in all three dimensions. Overall, the mean prediction error exhibited decreasing trends and converged to zero gradually.

The repeatability tests also reveal that the TV-based parabolic trajectory regulation shows better consistency and preciseness of

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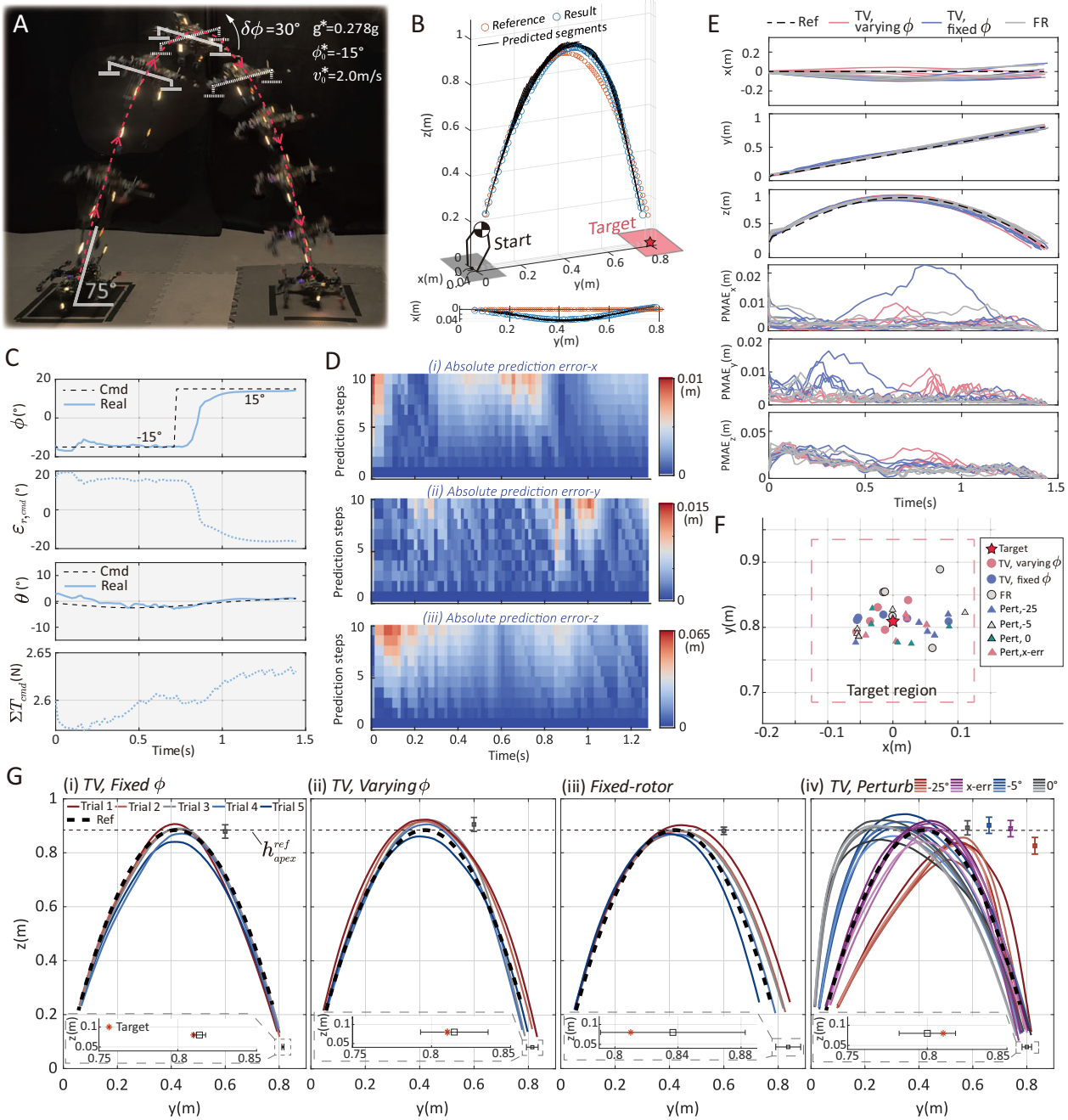


Fig. 5 | Targeted leaping onto confined region. A Leaping onto the robot's projection-sized region by employing the thrust-vectoring (TV) based MPC. In this test, ϕ is actively controlled and varied. The tilt thrust adapts to the change of ϕ while regulating the longitudinal trajectory. **B** Comparison of the reference, result and the predictions. **C** The controller commands and attitudes during the aerial phase. The thrust-vectoring angle command ε_r use $-\phi$ as the reference value to ensure the resultant thrust is closely along the vertical direction, which retain trajectories' parabolic characteristics. **D** The absolute prediction errors for individual steps in every time frame. The values of the prediction errors are indicated by the color bar. **E** Repeatability tests demonstrating tracking performance and prediction accuracy. Each group was conducted with five repeated trials. **F** Statistics of the exact landing points on the target region. The statistics include both the jumps under nominal conditions and the four perturbed cases (launch with inclination angle deviations and x-direction speed deviation). **G** Leap trajectories in the y-z plane. (i) Thrust-vectoring-based jumps with ϕ commanded to 0. (ii) Thrust-vectoring-based jumps with ϕ commanded from −15° to 15°. (iii). Fixed-rotor-based jumps. (iv) Robustness test investigating the performance of TV-based MPC on handling large takeoff angle deviations. For all trials, ϕ is commanded at corresponding launch angle (i.e., 0°, −5°, −15°, −25°) in the ascending phase and at −15° in the descending phase. The comparison of the predicted trajectories and experimental trials of (i)-(iv) is presented in Fig. S23 (nominal takeoff condition) and Fig. S24 (perturbed takeoff condition).

jumping distance, compared to the FR configuration. Among the three groups, TV mode with fixed ϕ demonstrated the best result in target landing (see Fig. 5G(i)), attaining a mean distance error of 3.8 mm with minimal variation. The jumping distance standard deviation (JDSD) reached 3.901 mm, an order of improvement compared to the FR case (4.589 cm), and almost a quarter of the record achieved by previous precise baselines such as Salto-1P (best record: 1.6 cm[47]) and Moobot (best record: 1.3 cm[19]), despite our jumper's considerably larger size.

Robust tests were conducted to analyze the TV-based controller's performance under takeoff condition deviations (see Fig. 5G(iv)). We

included four perturbed groups: three with takeoff angle deviations (−10°, +10°, and +15°) and one with lateral speed deviation (-0.2 m/s). All four groups employed the TV-based regulation with varying ϕ (launch with the corresponding pre-tilt angle, and land with a back tilt of +15°). Each group was tested over five trials, and all 20 tests confined the landing points within the target region (see Fig. 5F) with an average jumping distance error of 1 cm, and a JDSD of 1.77 cm. The controller commands for the jumps under nominal takeoff conditions are presented in Fig. S20, and robustness-test commands are in Figs. S21 and S22. The comparisons between MPC model predictions and experimental trajectories for each trial in Fig. 5G are shown in Figs. S23 and S24.

Bounce-free landings

Following a well-controlled parabolic trajectory, a stable robotic landing is the critical final step. The GravOff bipedal robot employed two landing methods according to targets' dimension and impact energy: passive damping (PD) and active damping (AD). The PD method utilizes the TPU shock absorber (see Fig. 1D) for jumps with high impact momentum, while AD is adopted for missions requiring a precise and bounce-free landing within confined targets.

We compared the proposed landing methods in two vertical jumping tasks, as illustrated in Fig. 6A, B. With AD, the robot maintained the legs extended in the aerial phase and employed the legs' synchronous contraction near the ground contact point as a soft deceleration mechanism (see Fig. 6A(i) and Supplementary Movie 8). In this process, the leg contraction speed was set at apex leap height and timing for flexion was based on a height threshold method (see supplementary section S9). The contraction movement of AD method extended ground contact duration (0.338 s) twofold compared to PD method (0.17 s), while reducing the average contact force ($\bar{F}_{AD} = m\bar{a}_{AD} = 0.367 \times 5.23 = 1.92N$; Fig. 6A(ii)) to a half of body weight and only a quarter of PD group ($\bar{F}_{PD} = m\bar{a}_{PD} = 0.367 \times 20.6 = 7.56N$; Fig. 6B(ii)). As a result, AD produced an impulse that almost dissipates the landing momentum: $I_{AD} = \bar{F}_{AD}\Delta T_{land} = 0.648N \cdot S \approx mv_{land}^{AD} = 0.69N \cdot S$, enabling the robot to remain stationary at the first contact point. The PD group saw an impulse ($I_{PD} = \bar{F}_{PD}\Delta T_{land} = 20.6 \times 0.367 \times 0.17 = 1.285N \cdot s$) that was 1.6 times its landing momentum, resulting in residual bounces deviating the robot 0.55 m away from the initial landing point. To initiate vertical jumps, both the AD and PD groups applied a leg motor rotational speed command of 800 deg/s with the attitude controlled horizontal.

In a jumping task onto target placed 1.35 meters away, the active damping method enabled the bounce-free landing without deviating out of the bounds (Fig. 6C, Supplementary Movie 8). The PD method, however, saw a deviation of about 0.2 m (see Supplementary Movie 8). After TCPT tracking, a back tilt identical to the launch angle was initiated at a height of 0.6 meters for the preparation of active damping. This ensured that the leg contact force and trajectory tangent at the landing point were aligned, preventing unstable flipping and falling upon contact.

Leap against wind disturbances

Robustness to disturbances is a critical factor in tracking problems, yet it is rarely explored in the context of jumping, especially during the aerial phase when disturbances like wind can affect landing position. In a targeted jumping task, we investigated our robot's response to wind disturbances generated by an electric fan, measured at 3.5 m/s using an anemometer (model RA110, Dongguan SANLIANG Electronics Co., Ltd., China) placed 1 meter away. Tests were conducted under three scenarios: TV-based jumping with inclination control, FR-based jumping, and jumping with open-loop control (attitude maintains level), as shown in Fig. 7A and Supplementary Movie 9. Results indicate that wind disturbances significantly affected all groups near the apex of the trajectory. However, the controller-enabled groups compensated for displacement loss in the y-direction by decreasing the inclination angle ϕ (FR-based case) or the tilt angle ε_r(TV-based case), as demonstrated in Fig. 7B. Despite imperfect tracking of the TCPT during descent, both controllers allowed the robot to reach the 0.25 × 0.25-meter target region (landing points correspond to trajectory endpoints in Fig. 7C; data presented as mean ± SD; all landing positions are presented in Table S15). Although both methods enabled the robot to reach the target region, FR-based MPC caused deviation beyond the region after first contact due to necessary forward inclination demands compensating for trajectory errors; in contrast, TV-based MPC maintained final position proximity to target with proper inclination control, as compared in Supplementary Movie 9.

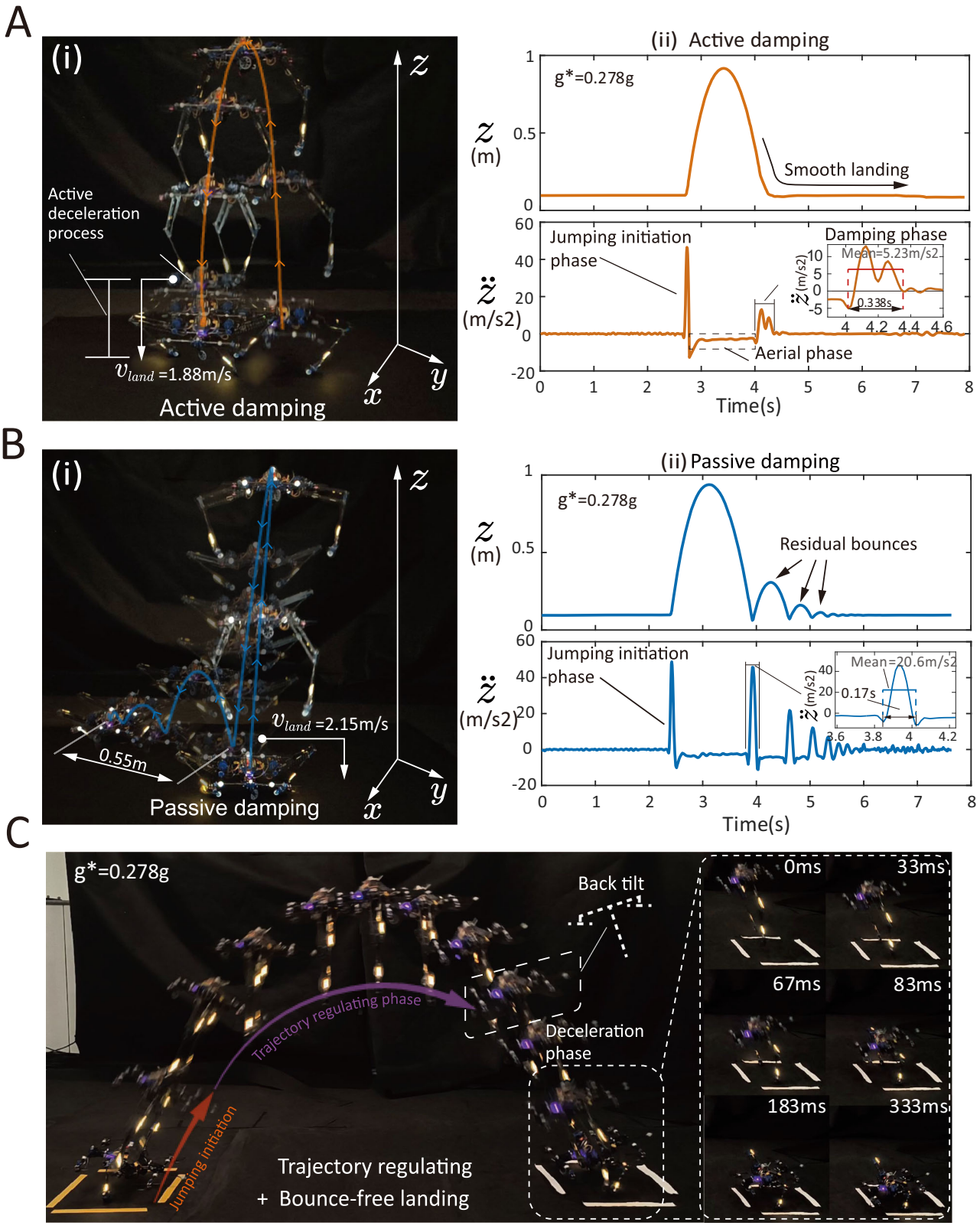
Continuous leaps across varying inclined targets

To exemplify the advantages of the TV configuration compared to the FR configuration, we challenged the GravOff robot with a demanding continuous jumping scenario featuring targets of varying surface angles (see Fig. 7D, Supplementary Movie 10). The inclined target platforms increased in height progressively from 0.6 m (first platform: 20° inclination, 0.3 m × 0.3 m) to 1.04 m (second platform: -25° inclination, 0.4 m × 0.4 m). In this task, the robot leaped and regulated three consecutive parabolas, traversed inclined targets A, B, before reaching the final target C. The parabolas have the nominal takeoff conditions set by $[\phi_0^* \quad v_0^* \quad g^*]$ and values set to $[-15^\circ \quad 2.05 \quad 0.278g]$, $[20^\circ \quad 2.25 \quad 0.278g]$, and $[-35^\circ \quad 2.15 \quad 0.278g]$, respectively. Once the robot landed on one inclined platform with legged assistive damping, the takeoff reference position was updated, with the previous first contact point as the start of the next reference parabola, as demonstrated in Fig. 7E. This task features the coordinated control of the inclination ϕ, tilt rotor angle ε_r, and the legs' damping and propulsion, in both the aerial and stance phases. While regulating the TCPT, the inclination angle was simultaneously controlled, ensuring adaptation to the inclined target upon the stance phase (controlled perpendicular to the target surface). The Corresponding control signals for the three components are shown in Fig. 7F. The damping signals were generated before the stance phase, with the motor speed set according to an estimate of the landing speed (see details in section S9). And the leg propulsion signals were generated once the legs reached full contraction with the speed set by equation (S23).

Predictive leaps through static/dynamic aerial obstacles

Since the tracking of parabolic trajectories is rigidly time-coupled, it indicates the timings for subsequent aerial positions ahead of a leap. Utilizing this feature, our robot showcased predictive jumps to tackle dynamic/static aerial obstacles. As shown in Fig. 8 and Supplementary Movie 11 (clip 1), we designed three scenarios where the robot was required to jump through a stationary or shifting window (0.51 by 0.35 m) with an error tolerance of 5 cm (lateral) and 10 cm (vertical). In the task of crossing the stationary window, the robot was required to follow the TCPT, adjust its leg pose to pass through the window without contact, and realize a soft landing (see Fig. 8A). The robot's legs experienced four phases: extension for jumping, contraction, second extension for damping preparation, and active damping. Figure 8B shows the controlled leap curve aligns closely with the reference, regulating the cross-point error to 1.4 cm (lateral) and 2.6 cm (vertical) for safe passage. The robot attained a 6.56 g peak acceleration and experienced a 0.278 g hypogravity, and softly landed with an average acceleration below g (Fig. 8C). The aerial phase attitude angle and corresponding controller commands are presented in Fig. 8D.

The second jumping task involves a continuously rotating window that hinders the robot from the targeted landing (see Fig. 8E, F and Supplementary Movie 11, clip 2). In this case, the robot must cross it. To ensure success, its Center of Mass (CoM) was expected to cross the



window plane at the time when the window plane was perpendicular to the positive y-axis (expected crossing position). With prior knowledge of the leap climbing time, including the time consumed in TCPT (0.56 s), acceleration (~0.1 s, estimated from statistical results, see Table S16), wireless communication and leg motor startup (~0.2 s; see Table S17), the robot made a predictive jump before the window reached the expected crossing position. The jump was initiated in advance at a phase angle of −84.3 degrees (0.86 s ahead), as shown in Fig. 8G, H. The actual CP (crossing position) deviated from the expected CP by 2.7 degrees. The exact time consumed in the climbing phase was 0.833 s, slightly smaller than the planned time of 0.86 s.

The third task exhibited jumping through the window in rapid pendular motion, as shown in Fig. 8I, J and Supplementary Movie 11 (clip 3). The GravOff robot jumped toward the direction in which the

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Fig. 6 | Active damping (AD) for stable bounce-free landing. A Vertical jumping and landing with active damping. (i) Images demonstrating the active damping. The GravOff robot utilized the legs' synchronous contraction to dissipate momentum during landing. The AD enabled the robot to maintain stationery without secondary bounces (ii) The height and vertical acceleration plot of the AD group. The peak acceleration attained in the acceleration phase was close to 5*g*. The average acceleration in the landing process was around half of Earth's gravity, within a duration of 0.338 s. The acceleration data were smoothed using the LOESS (Locally Estimated Scatterplot Smoothing) filter in Matlab, with the parameter (span) set to 0.05. **B** Vertical jumping and landing with Passive Damping (PD). Both active and passive damping tests were conducted using a leg motor takeoff speed of 800°/s. (i) Images demonstrating passive damping. The result indicates that the final landing point deviated by 0.55 meters from the initial landing point, after three obvious secondary bounces. (ii) The height and acceleration of the PD group. The impact acceleration reached a value more than four times that in the AD group, while the duration was half of the number in the AD group. Multiple impact acceleration spikes have been seen after the first landing. **C** Bounce-free landing on the targeted region. The result showed that the robot was capable of landing to a still in the 0.25-by-0.25-meter square region without secondary bounces. This illustrated the superiority of the AD compared to PD in scenarios with confined areas, as also can be seen in a comparison in Supplementary Movie 8.

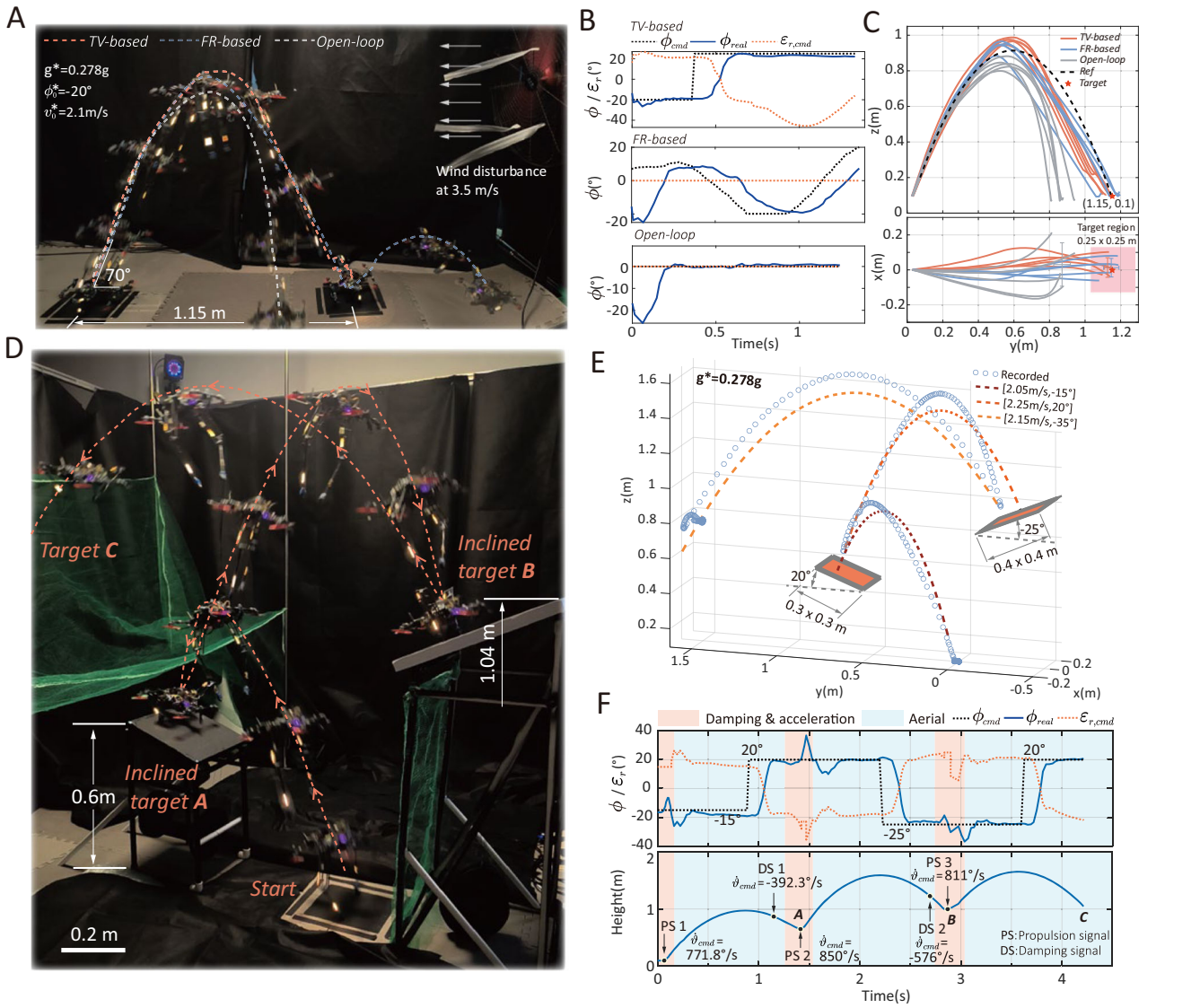
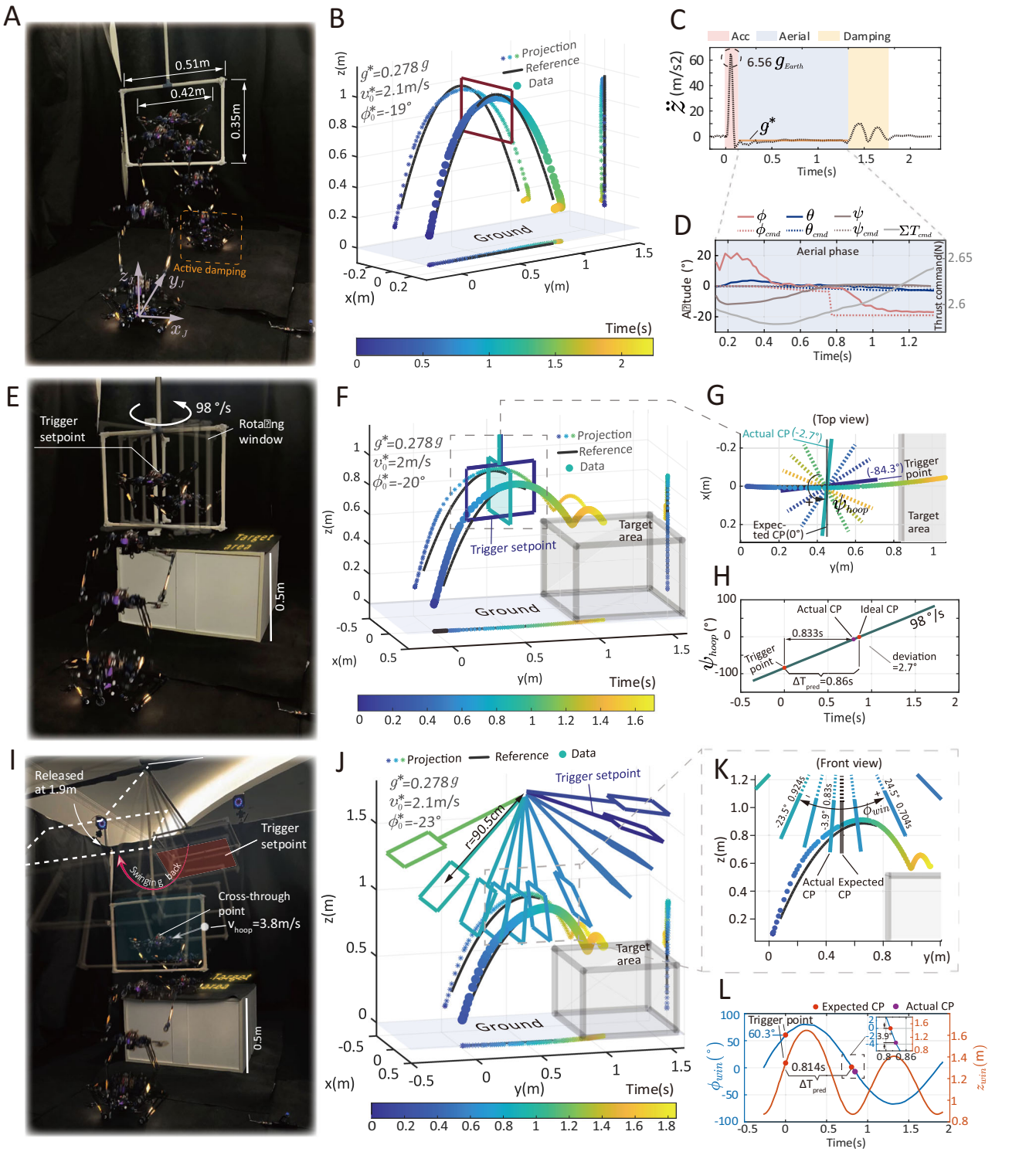


Fig. 7 | Leap missions exemplifying the advantage of thrust-vectoring (TV) based jumping trajectory regulation. A Comparison of jumping trajectories using vector-thrust, fixed-rotor (FR) based MPC and open-loop control. The trajectories were captured by video analysis tool Tracker (version 6.1.7). The wind disturbance (-3.5 m/s, measured by an anemometer 1 meter away) was generated by an electric fan directed opposite to the longitudinal motion of the TCPT trajectory. The MPCs (y-direction weight was set to 6 for FR-based MPC, and 5.5 for TV-based MPC) responded more aggressively near the jumping apex to compensate for longitudinal trajectory errors, via reducing the inclination (FR-based) or tilt (TV-based) angle, enabling landing onto the desired region. In contrast, open-loop control exhibited a significant wind blocking effect. TV-based MPC maintained the robot within the target region due to decoupled inclination control, achieving a proper landing contact angle (set to 25°). Conversely, trajectory correction of the FR-based group caused the robot to land with a forward inclination angle, deviating it outside the target bounds. **B** Inclination control response to the wind disturbance. The three plots correspond to the inclination and rotor tilt commands of the three trajectories in (**A**). **C** Repeatability tests of leaps against wind disturbance. Each group underwent five trials. Both FR-based MPC and TV-based MPC reached the target region in all trials, but TV-based MPC showed lower variation in jumping distance (JDSD: 1.85 cm) compared to FR-based MPC (JDSD: 4.33 cm). **D** A continuous jumping task across varying inclined targets. The GravOff robot conducted three successive leaps onto target A (20°; 0.3 m by 0.3 m), B (−25°; 0.4 m by 0.4 m) before reaching target C. **E** Recorded leap path by motion capture system and the references. **F** The coordinated controlled rotor tilt, inclination and leg actuation signals during the entire time sequence.

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pendular window swung and crossed through it before landing on the targeted area. The motion of the window was determined by a releasing height of 1.9 m (geometrical center height), resulting in a peak speed of 3.8 m/s at the crossing position. As the second group, the trigger timing was also determined by the climbing time of TCPT and the motion pattern of the aerial barrier. Specifically, the jumping signal was generated by the time when $\phi_{win} = 60.3°$ (Fig. 8K, L), which was 0.814 s before the expected CP. This task highlights an increased precision requirement for leap timing and TCPT tracking, where a mere 0.1 s delay (leading to a deviation of around 20 degrees) from the expected CP can lead to a collision or failure to pass through the window, as revealed by Fig. 8K. In the experiment, a 3.9-degree deviation (captured by the indoor positioning system) between the actual and expected CP.

Consecutive jumping missions involving moving, confined targets

To showcase the predictive feature of future position in TCPT-tracking-based jumping, we included another type of jumping experiment in which the landing target position varied with time. Specifically, we introduced a rotational motion on a square landing region (a 0.33-by-0.3-meter platform driven by a speed control servo)

Fig. 8 | Predictive leaping through aerial static and dynamic obstacles. **A** Consecutive images exhibiting jumping, narrow window crossing, and active damping (AD). In this task, the robot applied TCPT tracking to ensure safe passage through the window. The contraction of the legs (before crossing) was applied in case of collision. Active damping was used for a stable soft landing. **B** The jumping process data points and the reference trajectory. The color bar represents the elapsed time of each data point. Both the robot and window's position and attitude were tracked by motion capture system. **C** Robot's acceleration in the jumping process. **D** The attitude angle and controller commands in the aerial phase. The robot adopted a back tilt for preparation of inclined landing with AD. **E** Jumping through a rotating window before landing on a targeted area. A constant rotating speed of 98 deg/s was applied for the window (driven by a speed-control servo). The robot jumped through it by considering both the leap climbing time (including time for motor start-up, signal transfer, acceleration, aerial trajectory, 0.86 s in total) and the window's rotational speed, and selected leap timing to ensure safe passage. **F** The jumping data points, reference trajectories, and the window positions. **G** Demonstration of the deviation of the actual crossing position (CP) and expected CP. The crossing position is defined at the point where the y-coordinate of the robot's CoM and the geometrical center of the window align. **H** Recorded angle of rotating window. **I** Jumping through a pendular window. In the task, the pendular window was released at a constant height of 1.9 m, indicating its determined motion patterns with time. The robot jumped toward the window direction while the window (measured 3.8 m/s at the bottom) swung back. **J** Demonstration of the position and time where the obstacle crossing occurred, and comparison between data points and the reference. **K** Illustration of the actual CP and expected CP in front view. **L** Recorded angle and height of the pendular window.

slightly broader than the robot's foot span. Figure 9A demonstrates this scenario. The robot initiated a jump, followed the desired TCPT, employed the active-damping technique to land on the moving target, and seamlessly performed a second jump to land on a safety net, reaching apex heights of 0.95 m and 1.44 m in the two jumps. The jumping tasks experienced three acceleration spikes and two aerial hypogravity phases (g^*=0.264 g), shown in Fig. 9B. To land on the moving target, trigger timing was calculated according to the elapsed time in the reference TCPT (1.302 s), acceleration(~0.1 s), signal transfer and motor start-up (~0.2 s), and the rotational speed of the target region. We implemented this task with three rotational speeds for the platform: 33°/s, 76°/s and 93°/s (Supplementary Movie 12, clip 1). A similar time-related representation as Fig. 8 has been used to demonstrate the two ballistic curves of this jumping task (see Fig. 9C). In the group with $\omega_{platform} = 93^\circ/s$, the trigger setpoint corresponded to the position where $\psi_{platform} = 1.602 \times 93 \approx 149^\circ$($0^\circ$ corresponds to the position where the radius of the platform is perpendicular to the y-axis of $\boldsymbol{Q}_J$).

We conducted a repeatability test including ten trials using the group with $\omega_{platform} = 76^\circ/s$, as shown in Fig. 9D. Seven out of ten trials successfully landed and initiated a second jump on the rotating platform, and the remaining three failed trials revealed leg slippage due to late landing or bad tracking of TCPT (see Supplementary Movie 12, clip 2). The angular position of the platform $\psi_{platform}$ is presented in Fig. 9E, and the trigger setpoints were chosen by counting a predictive time of 1.602 s ahead of the expected landing point (0°). The landing points (where the z-direction acceleration turned positive from a negative value in the aerial phase) were statistically studied, with the average x and y location ± SD, as shown in Fig. 9F. Upon contact, the result reveals that the mean value of the angular position (−2.7°) was close to the expected value (0°), shown in Fig. 9G and Table S18.

Versatile locomotion in field tests

In addition to indoor TCPT tracking-based jumping, we evaluated the GravOff bipedal robot in field settings by integrating its three locomotion modes: air-propelled passive wheeling, synchronous legged crawling, and jumping (Fig. 10). Figure 10A depicts transitions between these modes. The legged crawling posture serves as an intermediate state for jumping and wheeling (see supplementary section S12 for mode transformation and crawling control) and proved effective as an alternative to jumping or hopping on terrain with unpredictable surface contact angles, such as grassland (Fig. 10B, Supplementary Movie 13), sand (Fig. 10C), cobbles (Fig. 10D), and grooved roads (Fig. 10E). The crawling motion enables minor turns and incremental steps, facilitating precise launch positioning and orientation adjustments during field-based jumping tasks.

For speed and efficiency, the air-propelled passive wheeling was adopted for locomotion on flat terrain. A speed test in (Fig. 10F, Supplementary Movie 14) revealed that the robot could attain a peak speed of 4.8 m/s (around 24 body lengths per second), with a 50% throttle command.

The hypogravity jumping strategy facilitates rapid traversal of large outdoor barriers. As shown in Fig. 10G, the robot successfully leaped over a 2.35 m-high wall under quasi-moon gravity, achieving a 3.3-meter apex height. To reach target locations, the robot executed locomotion mode transformations to navigate different types of terrains. This was demonstrated in scenarios involving targets above a staircase (crawling-jumping transition in Fig. 10H; wheeling-jumping transition in Fig. 10I), atop a 1.6 m wall (Fig. 10J, crawling-jumping transition), and across a 3 m-wide stream (Fig. 10K, crawling-jumping transition), as shown in Supplementary Movie 15.

Discussions

In this work, we present the GravOff robot, a system designed to leap by parabolic trajectory regulation under thrust-induced hypogravity, realizing ultra-range, precise, predictable and stable leap motions. The proposed gravity offset strategy significantly boosts leap elevation to 6.9 meters (46 times body height) under microgravity level, overcoming the mechanical constraint of legged propulsion while reducing to 25 % the landing momentum of the value under Earth gravity for the same height. These qualities allow the robot to traverse large obstacles in field tests while preserving landing safety, such as leaping over a 2.35 m wall, multi-level stairs, a 3 m wide stream, etc. The robot can execute both longitudinal and lateral jumps with the pretilt and the actuation phase difference methods. With the pre-tilt takeoff, the robot performed parabolic trajectory regulation via its inherent thrust vectoring configuration. For targeted leaps, this method proved robust, maintaining high leap consistency even when facing large takeoff angle deviation (± 10°) and strong aerial-phase wind disturbance (3.5 m/s). To ensure landing stability, we implemented active-damping technique for bounce-free landings on confined targets and passive damping for high-impact landings. By leveraging the time-coupled nature of the tracked trajectories, the robot had prior knowledge of future positions (and the time) before executing jumps, enabling predictive leaps in dynamic environments. These include leaping through fast-moving (rotating, pendular) windows, onto a shifting yet confined target platform, and seamlessly initiating consecutive jumps. During field tests, GravOff robot also demonstrated multimodal motions combining hypogravity jumping, legged crawling, and wheeling to tackle composite terrains.

The leg mechanism is crucial for thrust-induced hypogravity jumping and versatile motions across outdoor composite terrains, in terms of (i) Energy efficiency: Hypogravity jumping over small-to-medium obstacles enables significantly lower total energy demand. Ideally, the 300 mAh LiPo battery (3.42 Wh) could support 217 jumps onto the table in Fig. 3C versus only 18 flights. (ii) Vertical ascending speed and efficiency: Direct legged propulsion enables instant establishment of takeoff speed compared to air thrust propulsion, enabling agile crossing of dynamic targets (Fig. 8); (iii) Landing stability: During

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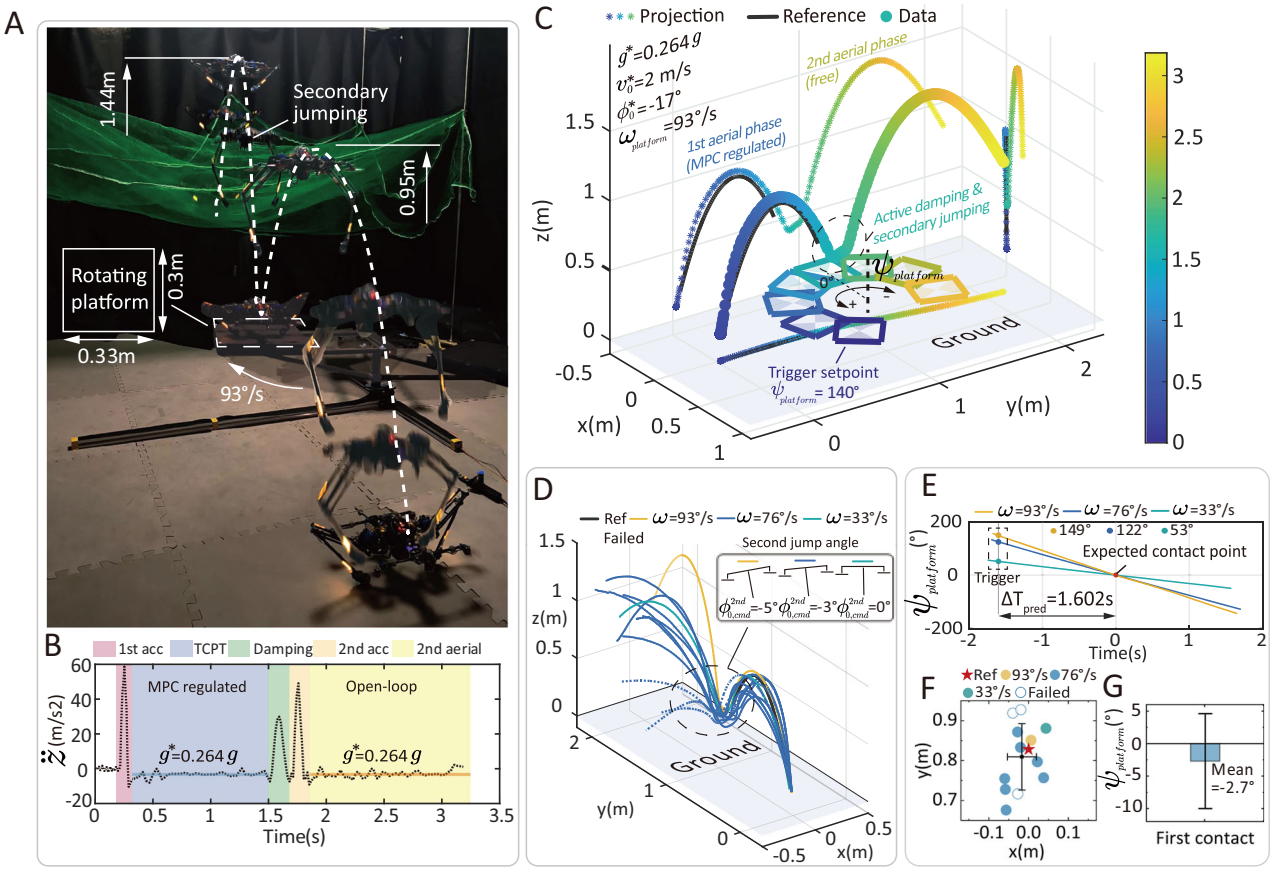


Fig. 9 | Consecutive jumping task with the target area in dynamic motions. A a consecutive jumping task with leap trajectory regulation, targeted landing on a rotating platform of 0.33 by 0.3 meters, active damping, and a secondary jump. The robot attained an apex height of 0.95 and 1.44 meters in the two jumps, respectively. **B** Z-acceleration during the jumping process (smoothed using the Matlab LOESS filter with a span of 0.1). An equivalent gravity of 0.264g was commanded in the two aerial phases. **C** Analysis on leaping trajectory and timing. The trajectory corresponds to the one presented in Fig. 8A. The reference is defined by the TCPT parameters (g^*=0.264 g, ϕ_0 = 17°, v_0 = 2 m/s). To ensure timely landing on the rotating platform, the robot was triggered 140° ahead of the expected landing position ($\psi_{platform}$ = 0°). **D** Jumping trajectories of all trials. The tasks were conducted across three distinct speed groups (33°/s, 76°/s, and 93°/s). A repeatability test was conducted in the 76°/s group, with a success rate of 70%. Bad tracking of TCPT and unprecise timing led to slippage in the failed groups, as also revealed by the detailed data in Table S18 and Supplementary Movie 12. In the second jump, different launch angles were applied for three groups according to the relative horizontal velocity of the robot to the platform. **E** Setting trigger position according to platform rotating speed. **F** Planar position of the first contact point on the platform. The position was taken at the point where the z-acceleration turned positive from a negative value, with Mean ± SD. **G** Platform angular position statistics upon first contact (Mean ± SD), see Table S18 for details.

the aerial descent phase, the legs perform active contraction (Fig. 4D) for bounce-free landings; (iv) Coordinated consecutive jumps: Coordinated control of leg extension and contraction, robot's attitude, and the vector mechanism ensures successful repetitive leaps in challenging missions (e.g., Fig. 7D, Fig. 9A). (v) Multimodal locomotion: jumping (as directed propulsion mechanism), crawling (initiating directional paddling-like motion), wheeling (as wheelable rigid supports).

We evaluate the leap performance of our work with a comparison (Table S20) to some existing works (from 0-1 kg) regarding leap precision level, height, consistency, landing stability, and other important functionalities as a jumper. To the best of our knowledge, rare research on robotic leaping can realize regulating parabolic trajectory and enabling the predictability of future discrete positions and timings of the aerial leap curve and landing point, before executing a jump. Compared to previous precise jumpers that make predictions by controlling launch angle and speed[19,47], the strategy of this work (launch speed/angle control with aerial regulation) alleviates the strict requirement on the takeoff condition precision and provides tolerance to post takeoff interferences (e.g., wind). Before this work, robotic leaps in an extraterrestrial low-gravity environment were still in the conceptual stage. The GravOff robot delves into this motion pattern under the Earth's environment, demonstrating advantages including boosted elevation and reduced landing impact.

Previous quadrotor or flapping-wing jumping hybrids[41,49–52] predominantly features fixed thrust configurations (thrust direction rigidly coupled to the airframe), employing open-loop control[41], non-parabolic position control[51], or touchdown angle and height regulation[49,50], or horizontal velocity and height control[52]. These works are yet to reach the precision level of regulating individual parabolas. In addition, when controlling desired attitudes, fixed-thrust-based jumpers also face additive translations (or unwanted attitudes when controlling translation). GravOff robot decoupled aerial inclination and parabolic trajectory control via synchronous thrust vectoring, which dynamically adapts the thrust direction to freely assigned inclination angles. TV-based regulation significantly improves jumping distance precision (Normalized mean distance error, see Table S20) and consistency (Coefficient of variation) over precise baselines like Salto-1P[47], Moobot[19], and other leg-only jumpers. The precision achieved can be attributed to superior responsiveness and accuracy

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Fig. 10 | Adapting to outdoor terrains via locomotion mode transition. **A** Illustration of the three locomotion modes and their applied scenarios. In synchronous legged crawling mode, the two legs perform paddling-like motions to pull the robot forward. The paddle span, depth, and frequency can be tuned (see supplementary section S12). By controlling the span differences in the two legs, the robot could follow a straight line or make turns. The crawling motion is suitable for irregular terrains where continuous jumping or hopping is inapplicable due to unpredictable foot contact angles. On flat terrain, the robot applied the air-propelled passive wheeling for speed and efficiency. Jumping was used to overcome large local barriers when paths to a target region were blocked during crawling or wheeling. **B** Legged crawling on deformable terrain (grassland). **C** legged crawling on deformable terrain (sand). **D** Legged crawling over a 0.1 m-high pebble pile. **E** Legged crawling on terrain with random grooves. **F** Air-propelled passive wheeling. **G** Jumping across a 2.35 m-high wall by utilizing the thrust-induced hypogravity condition. An apex height of 3.3 m was observed. **H** Combining crawling and jumping locomotion modes to overcome a seven-step stair. **I** Combining wheeling and jumping to overcome a four-step stair. The locomotion mode transformation process is illustrated in the inset figures. **J** Applying "crawling-jumping-crawling" to reach the desired position on a 1.6 m-high wall. **K** Jumping across a 3 m-wide stream.

offered by thrust vectoring control. Fixed thrust compensation relies on slower (> 0.2 s), oscillatory attitude rotation with steady-state errors, while thrust vectoring provides faster response, and negligible steady-state deviation (after calibration, see "Methods"). Although there exists TV-based flight control in previous study[63–65], there is no research focusing on jumping scenarios regulating sub-gravity projectile motion—a considerably faster scenario than flight, and achieved equivalent precision level. Compared to prominent drone-leg hybrid systems like Leonardo[66] and M4[67] that use flight for spatial obstacle clearance, GravOff's hypogravity leaping can achieve superior agility

and energy efficiency, with power consumption per unit mass ranging from 142 to 316 W/kg (equivalent gravity ranges from 0.8 to 0.1), less that of Leonardo (396.8 W/kg) and M4 (500 W/kg).

The distinguished functionalities of the hypogravity leap pattern may not be limited to the GravOff robot and can be generalized to most jumping robots. For example, the gravity offset strategy can be easily implemented on catapult-based robots, leveraging their high energy density (robots like[6–8,15]), to enable record-breaking jumps with safe landings. The TV-based dual regulation of the inclination and parabolic trajectory can serve as an effective reference for future thrust-assisted jumping hybrids for precise leap control in variational terrains.

Our current system relies on indoor positioning for spatial position feedback in trajectory regulation, is not available for outdoor settings. The field experiments conducted use the control of takeoff velocity, inclination, and gravity for parabolic trajectory generation, which sometimes revealed unwanted deviations as the contact surface varied. To address this, our next study aims to deploy TCPT tracking in field tests with onboard spatial position feedback (e.g., Lidar) and conduct computation on an embedded device.

In summary, our work unlocks the previously long-envisioned low-gravity leaping with a physical prototype, the GravOff robot, proving the superior advantages of this leap pattern within Earth's environment, and achieving parabolic trajectory regulation for precise, predictable, and deviation-tolerant leaps in dynamic settings. With its capability of simulating a wide range of gravity, the system can serve as an instructive design tool for legged jumping locomotion on other planets. Its agile and versatile performance may also suggest promising applications in rescue operations, such as traversing large debris, leaping into buildings with precision, or conducting predictive jumps to avoid environmental uncertainties.

Methods

Definitions of coordinates

The dynamic modeling of the proposed jumper considers a total of four coordinate frames, including the ground coordinate frame Q_G, the jumping coordinate Q_J, the robot's body coordinate Q_B, and the local rotor coordinate Q_r fixed to the front and rear rotor pairs (the two rotor pairs use the same coordinate as they have synchronized rotation angle), as demonstrated in Fig. 4B, C. The ground coordinate system Q_G is established in the motion capture system's calibration process. The jumping coordinate frame (Q_J) is defined at the robot's jump initiation position and orientation, with the z_J axis aligned with z_G axis and the y_J axis oriented towards the robot's current forward direction ahead of a jump. Q_B is located at the center of mass (COM) of the robot's main body and the y_B axis aligns with the rigid support frame. The rotor coordinate is fixed to each rotor pair and has the z-axis pointing perpendicular to the rotor rotation plane.

Ground contact phase dynamics

The ground contact phase dynamics constitute both the robot's acceleration phase and landing phase (including the active damping and passive damping). Before the launch of jumping, the front and rear rotor pairs have synchronized rotation angles and help to set at an arbitrary pre-tilt angle without slippage. In the acceleration phase, we assumed that the legs have a small linkage deformation in the lateral direction (i.e., along y_B, refer to Fig. 4A) and the extension trajectory and ground reactive forces are generated in the x_B-z_B plane. The ground contact phase dynamics can be described by Newton-Euler formalism, with the translational and rotational dynamics described in Q_J and Q_B, respectively:

$$\begin{cases} m\ddot{\mathbf{p}}_J = {}^J_B\mathbf{R}({}^B_r\mathbf{R}\mathbf{T}_{sum} + \mathbf{f}_{leg}) - mg\mathbf{e}_3 - \mathbf{f}_{drag} \\ \mathbf{J}_0\dot{\boldsymbol{\Theta}}_B = -\dot{\boldsymbol{\Theta}}_B \times \mathbf{J}_0\dot{\boldsymbol{\Theta}}_B + ({}^B_r\mathbf{R}\boldsymbol{\tau}_{thr} + \boldsymbol{\tau}_{leg}) - \mathbf{M}^B_\varepsilon \end{cases} \tag{3}$$

Here, $\mathbf{p}_J = [x_J, y_J, z_J]^T \in \mathbb{R}^{3\times1}$ is the position vector in Q_J, and $\boldsymbol{\Theta}_B = [\phi \quad \theta \quad \psi]^T$ denote the Euler angle. $\mathbf{T}_{sum} = [0 \quad 0 \quad \Sigma T_i]^T \in \mathbb{R}^{3\times1}$ is the resultant force produced by the four thrusters in Q_B, $\mathbf{f}_{drag} \in \mathbb{R}^{3\times1}$ corresponds to the aerodynamic drag, $\mathbf{M}^B_\varepsilon \in \mathbb{R}^{3\times1}$ is a moment vector representing the rotational drag of the main body and the gyroscopic effect of rotors. $\mathbf{e}_3 = [0 \quad 0 \quad 1]^T$ is the unit vector along the positive direction of axis z_J. ${}^J_B\mathbf{R}$ and ${}^B_r\mathbf{R}$ correspond to the rotation matrix from Q_B to Q_J and from Q_r to Q_B, respectively:

$${}^J_B\mathbf{R} = \begin{bmatrix} c\psi c\theta & c\psi s\theta s\phi - s\psi c\phi & c\phi s\theta c\psi + s\phi s\psi \\ s\psi c\theta & s\psi s\theta s\phi + c\phi c\psi & s\psi s\theta c\phi - c\psi s\phi \\ -s\theta & c\theta s\phi & c\theta c\phi \end{bmatrix} \tag{4}$$

$${}^B_r\mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c|\phi_0| & -s|\phi_0| \\ 0 & s|\phi_0| & c|\phi_0| \end{bmatrix} \tag{5}$$

where $c|\phi_0| = \cos|\phi_0|, s|\phi_0| = \sin|\phi_0|$. Note in the acceleration phase, the rotor rotation angle keeps the same value as the body inclination angle ϕ_0 (see Fig. 4A). Since the robot cannot be regarded as a rigid body considering the motion of the legs, the rotational dynamics consider only the main body and use $\mathbf{J}_0 \in \mathbb{R}^{3\times3}$ to refer to its inertia matrix. $\mathbf{f}_{leg} \in \mathbb{R}^{3\times1}$ and $\boldsymbol{\tau}_{leg} \in \mathbb{R}^{3\times1}$ are the force and torque exerting on the body from the left and right legs (The COM and the leg's force load points are separated by a distance d, see Fig. S3), defined in the body coordinate:

$$\mathbf{f}_{leg} = \begin{bmatrix} f^{Left}_{x_B} + f^{Right}_{x_B} & 0 & f^{Left}_{z_B} + f^{Right}_{z_B} \end{bmatrix}^T \tag{6}$$

$$\boldsymbol{\tau}_{leg} = \begin{bmatrix} 0 & f^{Left}_{z_B} d - f^{Right}_{z_B} d & 0 \end{bmatrix}^T \tag{7}$$

For longitudinal jumps (see Fig. 5), the two legs maintain synchronous extension (the same extension rate) to cancel out the horizontal component of the leg force and the body flipping torque, while the lateral jumps (Fig. 3) rely on the uncancelled f_{x_B} to produce the flip motion and lateral acceleration. And $\boldsymbol{\tau}_{thr} \in \mathbb{R}^{3\times1}$ is the torque generated by the thrusters in Q_r.

$$\boldsymbol{\tau}_{thr} = \begin{bmatrix} k_F(\omega_1^2 + \omega_4^2 - \omega_2^2 - \omega_3^2) \cdot l\, cos(\varepsilon_r)/2 \\ k_F(\omega_1^2 + \omega_3^2 - \omega_2^2 - \omega_4^2) \cdot w/2 \\ k_\tau(\omega_3^2 + \omega_4^2 - \omega_{r,1}^2 - \omega_{r,2}^2) \end{bmatrix} \tag{8}$$

Where $k_F\omega_i^2$ and $k_\tau\omega_i^2$ correspond to the produced thrust force and negative reactive torque of the ith thruster rotating at ω_i.

Aerial-phase dynamics

Once the jump is initiated, the robot encounters the aerial phase resembling a projectile motion under a gravity lower than the Earth's, as the resultant thrust force is maintained quasi-constant. And to ensure the compensated part of gravity is nearly constant, the rotation angles of the two rotor pairs are also synchronized (at the same ε_r). By eliminating the leg force and torque terms, the aerial-phase dynamics can be simplified to:

$$\begin{cases} m\ddot{\mathbf{p}}_J = {}^J_r\mathbf{R}\mathbf{T}_{sum} - mg\mathbf{e}_3 - \mathbf{f}_{drag} \\ \mathbf{J}\ddot{\boldsymbol{\Theta}}_B = -\dot{\boldsymbol{\Theta}}_B \times \mathbf{J}\dot{\boldsymbol{\Theta}}_B + {}^B_r\mathbf{R}\boldsymbol{\tau}_{thr} - \mathbf{M}^B_\varepsilon \end{cases} \tag{9}$$

Since there are small changes in the inertia properties led by the motion of legs in the aerial phase (thanks to the small mass of the carbon fiber linkages), we used a general form $\mathbf{J} = (J_{xx}, J_{yy}, J_{zz})_{diag} \in \mathbb{R}^{3\times1}$ to represent the inertia matrix of the complete robot. By replacing the takeoff angle ϕ_0 with an arbitrary rotor tilt angle ε_r, the expanded form

of dynamics is expressed by:

$$\begin{cases} \ddot{x}_J = \{\Sigma T[-s\varepsilon_r(c\psi s\theta s\phi - s\psi c\phi) + c\varepsilon_r(c\phi s\theta c\psi + s\phi s\psi)] - f_{drag,x}\}/m \\ \ddot{y}_J = \{\Sigma T[-s\varepsilon_r(s\psi s\theta s\phi + c\phi c\psi) + c\varepsilon_r(c\phi s\theta s\psi - s\phi c\psi)] - f_{drag,y}\}/m \\ \ddot{z}_J = [\Sigma T(-s\varepsilon_r c\theta s\phi + c\varepsilon_r c\theta c\phi) - mg - f_{drag,z}]/m \\ \ddot{\phi} = [(J_{yy} - J_{zz})\dot{\theta}\dot{\psi} + \tau_{x,thr} - M_{x,\varepsilon}]/J_{xx} \\ \ddot{\theta} = [(J_{zz} - J_{xx})\dot{\phi}\dot{\psi} + \tau_{y,thr}c\varepsilon_r - \tau_{z,thr}s\varepsilon_r - M_{y,\varepsilon}]/J_{yy} \\ \ddot{\psi} = [(J_{xx} - J_{yy})\dot{\phi}\dot{\theta} + \tau_{y,thr}s\varepsilon_r + \tau_{z,thr}c\varepsilon_r - M_{z,\varepsilon}]/J_{zz} \end{cases} \tag{10}$$

Coordinate transformation from Q_G to Q_J

Since trajectory regulation is conducted in the jumping coordinate frame $\boldsymbol{Q_J}$. To obtain the position and velocity in Q_J, a transformation from ground coordinate $\boldsymbol{Q_G}$ (where motion capture system data are obtained) to local jumping coordinate $\boldsymbol{Q_J}$ is performed:

$$\begin{cases} \mathbf{p}_J = {}^J_G\mathbf{R}\mathbf{p}_G - \mathbf{L}_G \\ \mathbf{v}_J = {}^J_G\mathbf{R}\mathbf{v}_G \end{cases} \tag{11}$$

where:

$${}^J_G\mathbf{R} = \begin{bmatrix} \cos(\psi_0) & \cos(\frac{\pi}{2} - \psi_0) & 0 \\ \cos(\frac{\pi}{2} + \psi_0) & \cos(\psi_0) & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{L}_G = [x_0^G \quad y_0^G \quad 0] \tag{12}$$

Where ψ_0 and (x_0^G, y_0^G) are the initial orientation and planar position prior to jumping, respectively.

Parabolic trajectory regulation with decoupled inclination control

To preserve ballistic characteristics in the jumping trajectory despite the tilt thrust, a method is required to maintain the resultant thrust vector as vertically as possible within the jump coordinate frame. Simultaneously, the tilt mechanism must smoothly modulate the trajectory in the forward direction. This can be realized by synchronizing the rotation angle of the tilt rotor (as the acceleration phase) and establishing a control reference that has $\varepsilon_r(t) + \phi(t) = 0$. To achieve smooth regulation of jumping trajectories rather than rigid tracking of TCPT at one time, we employed model-based predictive control (MPC) that could provide the anticipation of future states and allow for overall control of trajectory precision and smoothness via weights tuning.

The translational dynamics in Eq. (10) is utilized to establish the state-space models for the MPC. To address nonlinearities in (10) and for computational simplicity, the model is decomposed into planar (x-y) and vertical (z) dynamics:

$$\begin{cases} \dot{x}_J = u_J \\ \dot{y}_J = v_J \\ \dot{u}_J = \Sigma T[-s\varepsilon_r(c\psi s\theta s\phi - s\psi c\phi) + c\varepsilon_r(c\phi s\theta c\psi + s\phi s\psi)]/m - f_{drag,x}/m \\ \dot{v}_J = \Sigma T[-s\varepsilon_r(s\psi s\theta s\phi + c\phi c\psi) + c\varepsilon_r(c\phi s\theta s\psi - s\phi c\psi)]/m - f_{drag,y}/m \end{cases} \tag{13}$$

$$\begin{cases} \dot{z}_J = w_J \\ \dot{w}_J = \Sigma T(-s\varepsilon_r c\theta s\phi + c\varepsilon_r c\theta c\phi)/m - f_{drag,z}/m - g \end{cases} \tag{14}$$

Here $[u_J \quad v_J \quad w_J]^T$ corresponds to the velocity states. We assume the air drag is negligible in the planar (x-y) direction. The vertical drag $f_{drag,z}$, however, was found to be a noticeable disturbance term as the robot encounters much more vertical speed facing the thrusters (this is identical to the case where thrusters are under relative air flow), which alters the actual thrust generated. Here, $f_{drag,z}$ is modelled as a velocity-related damping term ($f_{drag,z} = c_d w_J, c_d = -0.0625$ based on the estimation from a vertical jump with constant thrust command, see details in section S23 and Fig. S28). In consideration of the speed-related force disturbance, the z-direction states-space model can be expressed by:

$$\begin{cases} \dot{\mathbf{x}}_z(t) = \mathbf{A}_z\mathbf{x}_z(t) + \mathbf{B}_z(t)U_z(t) - \mathbf{g} \\ \mathbf{y}_z(t) = \mathbf{C}_z\mathbf{x}_z(t) + \mathbf{D}_z U_z(t) \end{cases} \tag{15}$$

where:

$$\begin{cases} \mathbf{x}_z = [z_J \quad w_J]^T \\ U_z = \Sigma T \end{cases} \tag{16}$$

And,

$$\mathbf{A}_z = \begin{bmatrix} 0 & 1 \\ 0 & c_d/m \end{bmatrix}, \mathbf{B}_z(t) = \begin{bmatrix} 0 \\ \frac{-s\varepsilon_r(t)c\theta(t)s\phi(t) + c\varepsilon_r(t)c\theta(t)c\phi(t)}{m} \end{bmatrix}, \quad \mathbf{C}_z = \mathbf{I}_2, \mathbf{D}_z = \mathbf{0}, \mathbf{g} = [0 \quad g]^T \tag{17}$$

The x-y direction state-space model uses two compact terms ($\mathbf{U}_{xy} = [U_x \; U_y]^T$) as the control inputs to ensure the state space model is linear:

$$\begin{cases} \dot{\mathbf{x}}_{xy}(t) = \mathbf{A}_{xy}\mathbf{x}_{xy}(t) + \mathbf{B}_{xy}(t)\mathbf{U}_{xy}(t) \\ \mathbf{y}_{xy}(t) = \mathbf{C}_{xy}\mathbf{x}_{xy}(t) + \mathbf{D}_{xy}\mathbf{U}_{xy}(t) \end{cases} \tag{18}$$

where:

$$\begin{cases} \mathbf{x}_{xy} = [x_J \quad y_J \quad u_J \quad v_J]^T \\ \mathbf{U}_{xy} = \begin{bmatrix} -s\varepsilon_r(c\psi s\theta s\phi - s\psi c\phi) + c\varepsilon_r(c\phi s\theta c\psi + s\phi s\psi) \\ -s\varepsilon_r(s\psi s\theta s\phi + c\phi c\psi) + c\varepsilon_r(c\phi s\theta s\psi - s\phi c\psi) \end{bmatrix} \end{cases} \tag{19}$$

And,

$$\mathbf{A}_{xy} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_2 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}, \mathbf{B}_{xy}(t) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \Sigma T(t)/m & 0 \\ 0 & \Sigma T(t)/m \end{bmatrix}, \mathbf{C}_{xy} = \mathbf{I}_4, \mathbf{D}_{xy} = \mathbf{0}, \tag{20}$$

To eliminate the gravity term $\mathbf{g}$ in (15), we transform both the z and x-y direction translational dynamics (Eqs. (15) and (18)) into error dynamics form by subtracting both sides by the reference models:

$$\begin{cases} \begin{bmatrix} \dot{z}_{ref} \\ \dot{w}_{ref} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & c_d/m \end{bmatrix} \begin{bmatrix} z_{ref} \\ w_{ref} \end{bmatrix} + \begin{bmatrix} 0 \\ (-s\varepsilon_r c\theta s\phi + c\varepsilon_r c\theta s\phi)/m \end{bmatrix} \Sigma T_{ref} - \begin{bmatrix} 0 \\ g \end{bmatrix} + \begin{bmatrix} 0 \\ -c_d w_{ref}/m \end{bmatrix} \\ \begin{bmatrix} \dot{x}_{ref} \\ \dot{y}_{ref} \\ \dot{u}_{ref} \\ \dot{v}_{ref} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_2 \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} x_{ref} \\ y_{ref} \\ u_{ref} \\ v_{ref} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \Sigma T/m & 0 \\ 0 & \Sigma T/m \end{bmatrix} \begin{bmatrix} U_{x,ref} \\ U_{y,ref} \end{bmatrix} \end{cases} \tag{21}$$

which gives the error states and control inputs:

$$\begin{cases} \tilde{\mathbf{x}}_z = \mathbf{x}_z - \mathbf{x}_z^{ref} \\ \tilde{\mathbf{x}}_{xy} = \mathbf{x}_{xy} - \mathbf{x}_{xy}^{ref} \end{cases}, \begin{cases} \tilde{U}_z = U_z - U_z^{ref} \\ \tilde{\mathbf{U}}_{xy} = \mathbf{U}_{xy} - \mathbf{U}_{xy}^{ref} \end{cases} \tag{22}$$

Note, the z-direction reference control input is set according to the equivalent gravity of the TCPT ($U_{z,ref} = \Sigma T_{ref} = m(g - g^*)$), and x and y reference control inputs are both zeros ($U_x^{ref} = 0, U_y^{ref} = 0$), which is identical to the case with no attitude or the rotor tilt reactions. Using forward Euler method, the state space model is discretized to establish

the prediction models:

$$\begin{cases} \tilde{\mathbf{x}}_{xy}(k+1) = \dot{\tilde{\mathbf{x}}}_{xy}(k) \cdot \Delta t + \tilde{\mathbf{x}}_{xy}(k) \\ \tilde{\mathbf{x}}_{z}(k+1) = \dot{\tilde{\mathbf{x}}}_{z}(k) \cdot \Delta t + \tilde{\mathbf{x}}_{z}(k) \end{cases} \tag{23}$$

where k corresponds to the kth time frame and Δt is the sampling time.

Since motor vibrations also induce stochastic thrust fluctuations, as evidenced by the thrust-command profile in Fig. S9C. To capture this effect, we incorporate an unmeasured disturbance $d_n(k)$ modeled as a random walk noise:

$$d_n(k+1) = d_n(k) + a_n w_n(k) \tag{24}$$

where $w_n(k) \sim \mathrm{N}(0, 1)$ denotes the discrete-time white noise, and $a_n = 0.05$ is the assigned noise amplitude (clamped by 0.05 N). With the noise modeling, the complete prediction state-space prediction model in the z-direction is given by:

$$\dot{\tilde{\mathbf{x}}}_z(k+1) = \mathbf{A}_{d,z}\tilde{\mathbf{x}}_z(k) + \mathbf{B}_{d,z}(k)\tilde{U}_z(k) + \mathbf{d}_n(k) + \mathbf{d}_d(k) \tag{25}$$

with:

$$\mathbf{A}_{d,z} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 + \frac{c_d \Delta t}{m} \end{bmatrix}, \mathbf{B}_{d,z}(k) = \begin{bmatrix} 0 \\ \frac{\Delta t(-s\varepsilon_r(k)c\theta(k)s\phi(k) + c\varepsilon_r(k)c\theta(k)c\phi(k))}{m} \end{bmatrix}, \tag{26}$$

$$\mathbf{d}_n(k) = \mathbf{B}_{d,z}d_n(k)(\text{noise}), \mathbf{d}_d(k) = \begin{bmatrix} 0 \\ \frac{\Delta t c_d w_{ref}(k)}{m} \end{bmatrix}(\text{measurable disturbance}) \tag{27}$$

Subtracting the x-y reference in (20), the x-y direction error state-space prediction model is similarly derived:

$$\dot{\tilde{\mathbf{x}}}_{xy}(k+1) = \mathbf{A}_{d,xy}\tilde{\mathbf{x}}_{xy}(k) + \mathbf{B}_{d,xy}(k)\tilde{\mathbf{U}}_{xy}(k) \tag{28}$$

where,

$$\mathbf{A}_{d,xy} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \mathbf{B}_{d,xy}(k) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \Sigma T(k)/m & 0 \\ 0 & \Sigma T(k)/m \end{bmatrix} \tag{29}$$

With readings of the positions and velocities (x_J, y_J, z_J, u_J, v_J, w_J) transformed by Eq. (11) from global coordinates, the error states are obtained by subtracting the TCPT reference in (1), and passed as inputs to two adaptive MPC solvers in Matlab, while the attitude angles ($\phi(k), \theta(k)$) retrieved from the motion capture cameras and the previous rotor tilt angle $\varepsilon_r(k-1)$ update matrix $\mathbf{B}_{d,z}(k)$(as presented in the control workflow in Fig. 4D), and previous thrust command $\Sigma T(k-1)$ update the matrix $\mathbf{B}_{d,xy}(k)$, simultaneously.

In each step, the two applied controllers calculate the optimal control inputs $\tilde{U}_{xy}$, $\tilde{U}_z$ by minimizing the cost function J through solving a quadratic programming (QP) problem under assigned constraints:

$$\begin{aligned} \tilde{U}_i = \underset{U_i}{\mathbf{argmin}}\,|_{i\in\{xy,z\}} \Bigg\{ & \sum_{j=1}^{p} [\mathbf{y}_i(k+j|k) - \mathbf{y}_{i,ref}(k+j|k)]^T \mathbf{Q}_i [\mathbf{y}_i(k+j|k) - \mathbf{y}_{i,ref}(k+j|k)] \\ & + \sum_{j=0}^{p-1} [\tilde{U}_i(k+j|k) - \tilde{U}_{i,ref}(k+j|k)]^T \mathbf{R}_i [\tilde{U}_i(k+j|k) - \tilde{U}_{i,ref}(k+j|k)] \\ & + \sum_{j=0}^{p-1} [\tilde{U}_i(k+j|k) - \tilde{U}_i(k+j-1|k)]^T \mathbf{R}_{i,\Delta U} [\tilde{U}_i(k+j|k) - \tilde{U}_i(k+j-1|k)] \Bigg\} s.t. \\ & -\sin(\theta_{\min}) \le \tilde{U}_x \le \sin(\theta_{\max}), \; -\sin[(\varepsilon_r + \phi_{curr})_{\min}] \le \tilde{U}_y \le \sin[(\varepsilon_r + \phi_{curr})_{\max}], \\ & -1 \le \tilde{U}_z \le 2. \end{aligned} \tag{30}$$

here, $\mathbf{Q}_i, \mathbf{R}_i, \mathbf{R}_{i,\Delta U}$ correspond to the weighting matrices on the measured outputs $\mathbf{y}_i$, control inputs $\tilde{U}_i$, and the variational rate of control inputs, respectively, which are set to:

$$\begin{cases} \mathbf{Q}_{xy} = diag(2.9\ 2.5\ 0.2\ 0.5) \\ \mathbf{R}_{xy} = diag(1\ \ 1) \\ \mathbf{R}_{xy,\Delta U} = diag(0.3\ 0.3) \end{cases}, \begin{cases} \mathbf{Q}_z = diag(2.2\ 0) \\ \mathbf{R}_z = 1 \\ \mathbf{R}_{z,\Delta U} = 0.08 \end{cases} \tag{31}$$

Note, in the x-y direction, the controller is constraint by the allowable rotation range of $\theta(|\theta| \le 10^\circ)$ and the allowable relative rotation angle of tilt rotor ε_r with respective the negative value of current inclination angle ϕ_{curr}, i.e., $-\phi_{curr} - 30^\circ \le \varepsilon_r \le -\phi_{curr} + 30^\circ$.

The actual assignable attitude and rotor tilt commands (θ, ε_r) are obtained by solving:

$$-s\phi c\psi s\varepsilon_r s\theta + s\varepsilon_r s\psi c\phi + c\varepsilon_r c\psi s\theta c\phi + c\varepsilon_r s\phi s\psi = \tilde{U}_x \tag{32-a}$$

$$-s\phi s\psi s\varepsilon_r s\theta - s\varepsilon_r c\psi c\phi + c\varepsilon_r s\psi s\theta c\phi - c\varepsilon_r s\phi c\psi = \tilde{U}_y \tag{32-b}$$

By (32-a), having:

$$s\theta c\phi = \frac{\tilde{U}_x + s\phi c\psi s\varepsilon_r s\theta - s\varepsilon_r s\psi c\phi - c\varepsilon_r s\phi s\psi}{c\varepsilon_r c\psi} \tag{33}$$

Substituting (33) into (32-a), yield:

$$s\varepsilon_r\left(\frac{s^2\psi c\phi}{c\psi} + c\phi c\psi\right) + c\varepsilon_r\left(\frac{s^2\psi s\phi}{c\psi} + s\phi c\psi\right) = \frac{s\psi}{c\psi}\tilde{U}_x - \tilde{U}_y \tag{34}$$

Letting:

$$A = \frac{s^2\psi c\phi}{c\psi} + c\phi c\psi, B = \frac{s^2\psi s\phi}{c\psi} + s\phi c\psi, C = \frac{s\psi}{c\psi}\tilde{U}_x - \tilde{U}_y \tag{35}$$

where A, B and C are known values (ψ and ϕ are read from motion capture system, and $\tilde{U}_x$ and $\tilde{U}_y$ are obtained from controller) at each time step. (34) is hence reformatted into:

$$\sqrt{A^2 + B^2}\,\sin(\varepsilon_r + \varphi) = C, \varphi = \arctan\left(\frac{B}{A}\right) \tag{36}$$

which yields:

$$\varepsilon_r = \arcsin(C/\sqrt{A^2 + B^2}) - \varphi \tag{37}$$

The value of θ is yielded by substituting (37) back to (32-a):

$$\theta = \arcsin\left(\frac{\tilde{U}_x - s\varepsilon_r s\psi c\phi - c\varepsilon_r s\phi s\psi}{c\varepsilon_r c\phi c\psi - s\phi c\psi s\varepsilon_r}\right) \tag{38}$$

The rotor tilt value (37), roll angle (38), an actively-controlled inclination ϕ_{cmd}^{active}, yaw angle command ($\psi_{cmd}=0$), together with the thrust command ($\tilde{U}_z+\Sigma T_{ref}$) are assigned to low-level controllers, and the thrust vectoring servos, as demonstrated by the control workflow in Fig. 4D. Note, here the reference position of ε_r corresponds to the negative value of the inclination angle ($-\phi$), as:

$$\tilde{U}_y^{ref}\Big|_{\substack{\psi_{ref}=0\\ \varepsilon_{r,\,ref}=-\phi}} = s\phi c\phi - c\phi s\phi = 0 \tag{39}$$

For the case of FR-based configuration, the control DOF in y direction is taken by ϕ, and set rotor to be fixed ($\varepsilon_r=0$) in (32) to solve for θ and ϕ with known ψ.

Thruster tilt angle calibration

To establish precise tilt angle control, we calibrated the relationship between servo rotation angles and pulse-width modulation (PWM) input signals. A cross-shaped calibration bar with infrared reflective markers (Fig. S30) was rigidly mounted to the servo rotor. Using motion capture system (Swift 30 A, LUSTER LightTech Co., Ltd., China), we constructed the calibration bar as rigid body and tracked its angle values. We assigned step a command between −15° and 15° to the servo, and calibrated PWM value was determined when the measured rotation matched the commanded angle within a tolerance of ±0.5°.

Inclination requirement upon landing

To achieve stable landing within the target region using active damping, the robot needs to pre-regulate its body inclination angle during the aerial phase to prevent tumbling upon ground contact. Since the legs maintain fixed contact during landing, it can be modeled as a pin joint in the *y-z* plane (see an illustration in Fig. S29). The planar landing speed (in y-z plane) can be decomposed into the direction along the legs and the legs' tangential direction. The radial speed component is damped by the active damping force from the legs with a proper contraction speed (see details in section S9), governed by the impulse theorem:

$$\mathbf{e}_3\bar{\mathbf{f}}_{leg}\Delta T_{damp} = m\left\|\mathbf{v}_{yz}^{LD}\right\|\cos\,\delta\phi \tag{40}$$

where $\bar{\mathbf{f}}_{leg}$ and ΔT_{damp} denote the average leg force (in $\boldsymbol{Q}_B$) and the duration of landing phase, and $\delta\phi$ corresponds to the angle between the leg line (side-view) and landing velocity $\mathbf{v}_{yz}^{LD}$ (at the first contact point with ground). While the legs provide the damping forces, the tangent direction velocity causes the rotation around the pin joint:

$$\omega_{LD} = \frac{\left\|\mathbf{v}_{yz}^{LD}\right\|\sin\,\delta\phi}{r_{LD}} \tag{41}$$

where r_{LD} denotes the radius of rotation (from COM to the pin joint) at the first contact point of landing. Although the rotation can be partly suppressed by the thrust that controls the attitude, the reduction of the radius $r(t)$ also leads to the decrease of inertia moment, causing an unproportional increase of the rotation speed:

$$J_{LD}\omega_{LD} = J(t)\omega(t) \Rightarrow mr_{LD}^2\omega_{LD} = mr^2(t)\omega(t) \Rightarrow \omega(t) = \omega_{LD}\frac{r_{LD}^2}{r^2(t)} \tag{42}$$

In this sense, to avoid the tangent velocity, the attitude of the robot is adjusted such that:

$$\phi_{LD} = \arctan(v_{y,LD}/v_{z,LD}) \tag{43}$$

Robot fabrication

The leg actuator uses a high KV brushless DC motor (T Motor F1408 3950KV, Tiger Motor International Co., Ltd, China) reduced by a four-stage steel gear set (teeth numbers: 11; 25/10; 35/8; 36/14; 44, reduction ratio 112.5) manufactured using powder metallurgy (compressing steel powder in a mold and sintered between 1120 and 1250 °C). The gears are sandwiched between two 1.5 mm-thick CNC-machined (using a C6090 router machine, Wuhan Baike Times Laser Technology Co., Ltd) carbon fiber composite plates linked by four nylon M2 threaded hex spacers. The gears use axial stoppers (3D-printed with ESUN hard tough resin, Shenzhen Esun Industrial Co. Ltd, using Elegoo Saturn 2, Shenzhen Elegoo Technology Co., Ltd) to ensure engagement. To read the angle position using the magnetic encoder (As5600, Ams-OSRAM AG), a radial ring magnet (D7-d3-h3) is mounted to the other side of the output shaft (Fig. 1D(ii)), and rotated to the calibrated position (180 degrees) before being super-glued (using ergo 5210 adhesive, Kisling AG). A steel-gear servo motor (DS1906 B, GDW Digital Servo Manufacturer Co., Ltd, China) is mounted beside (in opposite direction) the brushless motor by a 3D-printed fixer (see Fig. 1D(ii)). The output shaft of the servo and a coaxial steel shaft (a two-head threaded shaft in 1.5 mm, with one end inserted into the gearbox) fix the whole leg actuator to a 3D-printed hip fixer (polyethylene terephthalate glycol (PETG), printed with Bambu Lab A1 mini, Shenzhen TuoZhu Technology C0., Ltd, China). The leg upper links (1.5 mm thick, Yangzhou Lihong Carbon Fiber Materials Co., Ltd) were also CNC machined and were mounted with flanged ball bearings (F681X, NSK Ltd.) at the joints, and linked by a 3D-printed two-axis connector (Fig. 1D(i)) to the lower link (a D5-d4 mm hollow carbon fiber tube). The TPU impact absorber was 3D-printed (TPU95A-HF, FusRock Co., Ltd) and mounted between the two linkages of the upper leg.

The brake mechanism is driven by a coreless motor (RIC-615D, Ric Motor Co., Ltd, China) reduced by a gear reduction set (two stages: 33/8-33/9) and a small screw slider mechanism. The final-stage gear drives the rotation of a screw rod (M1.6 × 0.35) and a hex nut (also M1.6 × 0.35) mounted into the 3D-printed slider (3D-printed with UV resin) transfer rotation into linear motion. A slice of 400-grit sandpaper was glued to the brake for increased friction. The thrust vectoring unit uses a pair of 1:1 angle gear (14 teeth, 3D-printed UV resin), with the output gear fixed to a hollow carbon fiber tube (D5-d4) using a tapping screw. At the two ends of the tube, the thrusters (GTS V2 1202.5 6000KV, RCinpower Co., Ltd, China) were mounted in the same horizontal plane.

Robot calibration

The robot's setup requires the calibration of the leg motor, leg servos, and the pitch turning servos. For the leg motor, once the radial magnet is glued, its angular position is fixed. We included a correction factor and manually tuned it in our code to eliminate the remaining errors from the calibration position (see Fig. S27A). Since the applied PWM (Pulse Width Modulation) servos have no position readback, we first calibrated their rotation ranges to 180 degrees in the code and obtained a mapping between PWM value and actual angle (measured by an as5600 magnetic encoder module) and then set the zero-degree position (Fig. S27A). The servo angles are then directly controlled by setting the corresponding PWM values.

Indoor positioning setup

The robot tracking was conducted in a 3.4 m × 3.2 m × 2.2 m arena, with six evenly spaced motion capture cameras (SWIFT 30, LUSTER LightTech Co., Ltd., China) mounted on a cubic aluminum frame. The angle of the cameras was adjusted to ensure the experimental space was in the cameras' field of view (83° × 66° for each camera). Before conducting TCPT tracking test, the cameras were calibrated using a 250 mm wand and the world coordinate frame and levelness of ground

were set and tuned using a calibration triangle with an embedded level vial.

Procedures for leaping in dynamic environments

Infrared markers were attached to the robot (7 markers), the window (5 markers), and the rotational target platform (5 markers) to construct rigid bodies. The robot's CoM was calculated by SolidWorks (version 2019, Dassault Systèmes) and was set in the motion capture software Moca 2.3.0.8. Their position and attitude data were streamed to Matlab (2019b, MathWorks, Inc.) for real-time leap control. Before leaping, the robot was placed at the launch point by referring to the initial point of the tracked trajectory. The leap tasks calculated the exact trigger points. During implementation, since the streamed data are discrete, the experimental trigger timings were set to be the first data points surpassing the calculated timing (Figs. 7H, L, 8E). In the window-crossing tasks, the contraction and extension of the legs in the aerial phase were controlled according to the current elapsed time in TCPT, which reflects the robot's spatial position. For tasks in Fig. 7, considering the motor startup time and signal transfer (~200 ms) and contraction process(~160 ms), the contraction signals were generated 400 ms before reaching the window to ensure safe passage. The leg extension signal (only for task in Fig. 7H) was generated 70 ms (~5 MPC loops) after window crossing. For the task in Fig. 9, the legs first adopted a slow contraction signal before reaching the height threshold for damping to eliminate preparation time for motor start-up (see supplementary section S9 for details). The leg command signals, thrust, pitch and roll angle commands were all generated in Matlab and transmitted wirelessly to the development board Teensy (Teensy® 4.0) via two HC-12 modules (transmitter and receiver, Guangzhou Huicheng Information Technology Co., Ltd, China) to control leg movement and attitude.

Data availability

All the data generated in this study to support the paper's conclusions are presented in the main manuscript and supplementary information.

Code availability

The code for robot control in this study (including the parabolic trajectory regulation and the code used in embedded device Teensy) has been deposited in public repository GitHub https://github.com/Zijiesun1236/GravOff-robot-/tree/main.

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Author contributions

Y.M. LI and L. TENG co-supervised this research. Z.J. SUN and J.G. ZHAO co-developed the methodologies of using synchronous thrust vectoring for parabolic trajectory regulation. L.TENG proposed the multimodality for the robot. Z.J. SUN designed, fabricated the robot prototype. Z.J. SUN and J.G. ZHAO developed the programs for control, constructed the experimental setup, and conducted the robot tests. Z.J. SUN processed and analyzed the data and completed the visualization of the results. Z.J. SUN, J.G. ZHAO, Y. M. LI, L. TENG accomplished manuscript writing and editing. L. TENG and Y.M. LI provided funding support for this research.

Competing interests

The authors declare no competing interests.

Additional information

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